



US 20250285165A1

(19) **United States**

(12) **Patent Application Publication**
Schack et al.

(10) **Pub. No.: US 2025/0285165 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **METHOD, COMPUTER PROGRAM
PRODUCT, AND SYSTEM FOR DISPLAYING
AUGMENTED REALITY ELEMENTS FOR
NAVIGATING TO A LOCATION OF AN ITEM
WITHIN A WAREHOUSE**

Publication Classification

(51) **Int. Cl.**
G06Q 30/0601 (2023.01)
G06Q 10/087 (2023.01)
G06T 19/00 (2011.01)
(52) **U.S. Cl.**
CPC *G06Q 30/0631* (2013.01); *G06Q 10/087*
(2013.01); *G06Q 30/0633* (2013.01); *G06Q*
30/0643 (2013.01); *G06T 19/003* (2013.01);
G06T 19/006 (2013.01)

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(21) Appl. No.: **19/216,641**

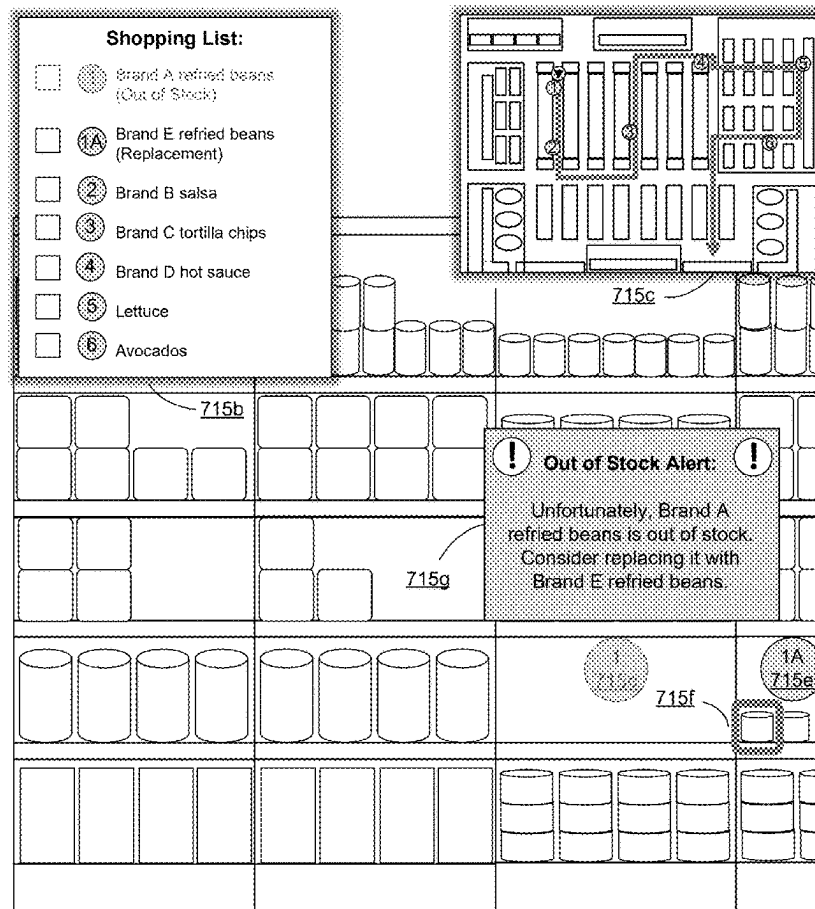
(22) Filed: **May 22, 2025**

Related U.S. Application Data

(63) Continuation of application No. 18/236,370, filed on
Aug. 21, 2023, now Pat. No. 12,333,592, which is a
continuation of application No. 17/877,718, filed on
Jul. 29, 2022, now Pat. No. 11,783,400.

(57) **ABSTRACT**

A wayfinding application executing on a client device receives a current location of the device within a warehouse and accesses a layout of the warehouse describing locations of items included among an inventory of the warehouse. The application identifies a route from the current location to one or more target items, generates augmented reality elements including instructions for navigating the route, and sends the elements to a display area of the device. The application detects a location within the warehouse associated with a target item and determines whether the item is at the location based on an image captured by the device. Upon determining it is not at the location, the application alerts a user of the device to a replacement item by generating an augmented reality element that calls attention to it and sending the element to the display area.



100

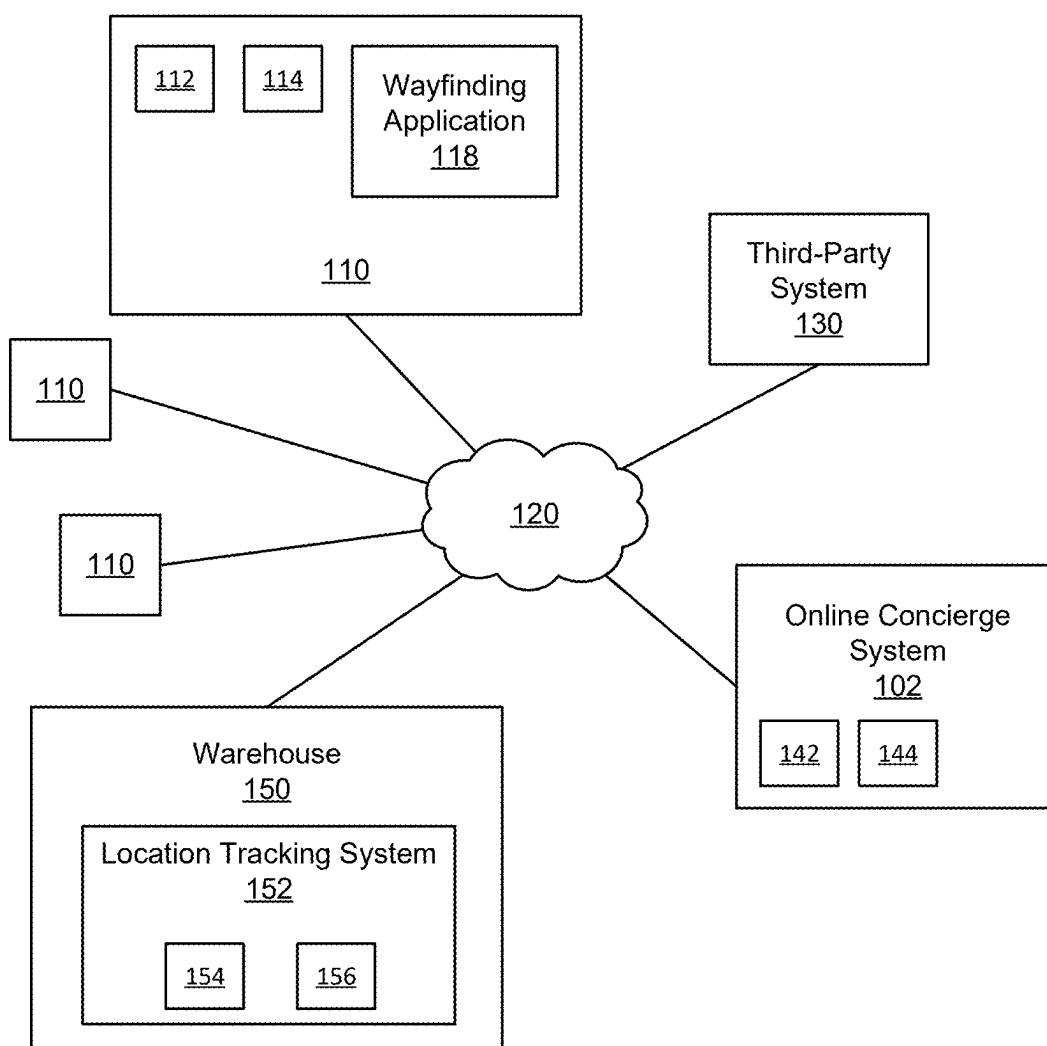


FIG. 1

200

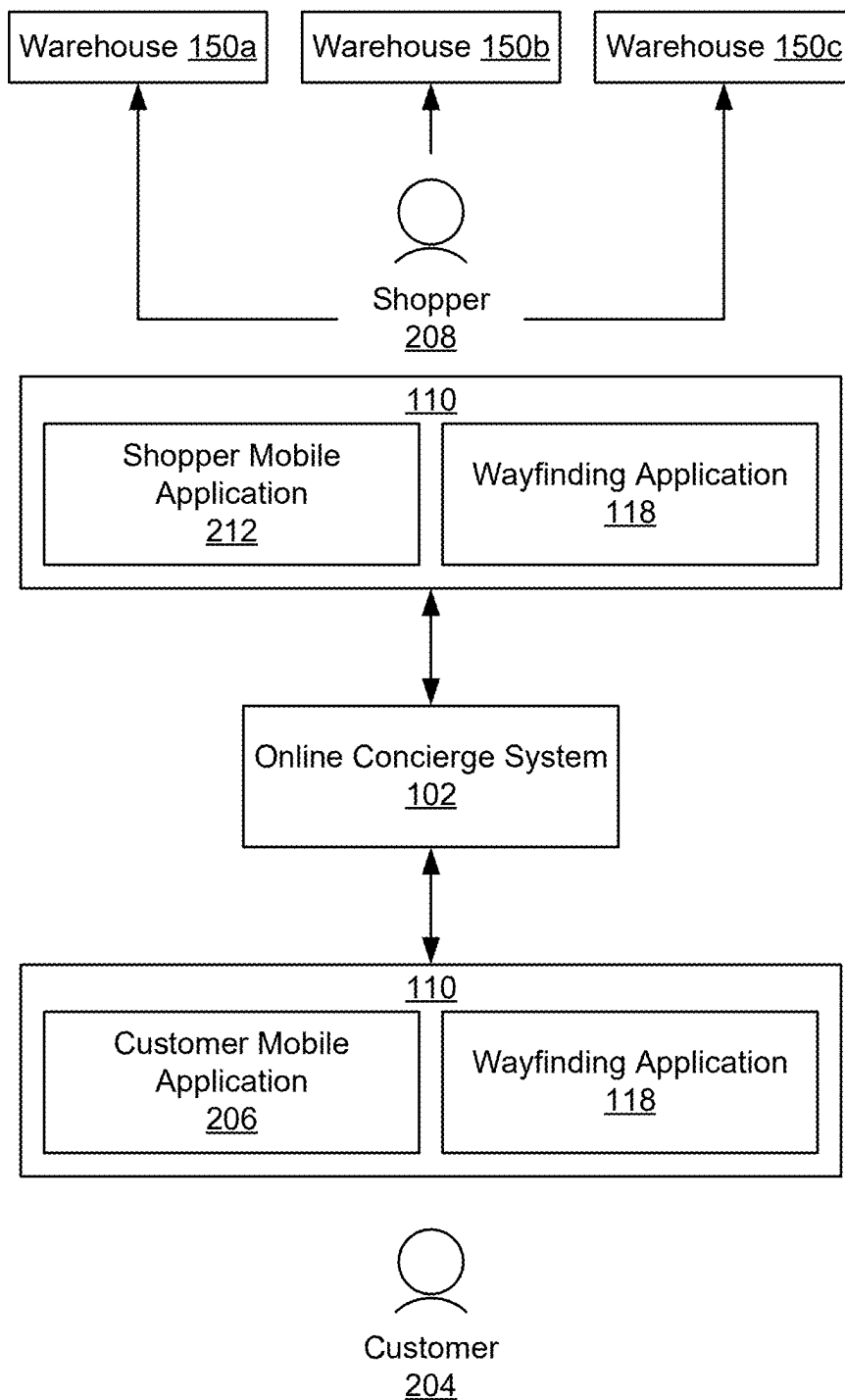


FIG. 2

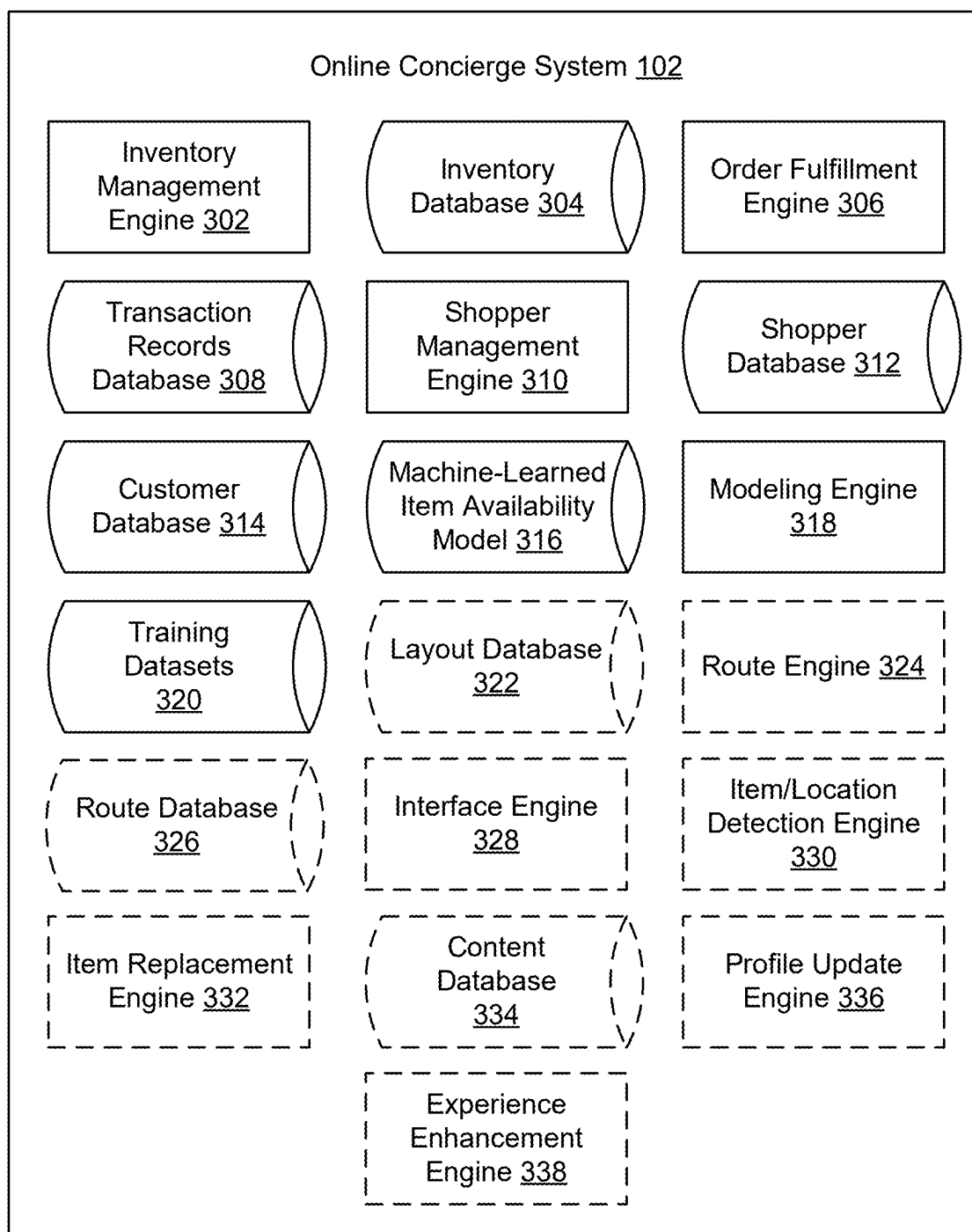


FIG. 3A

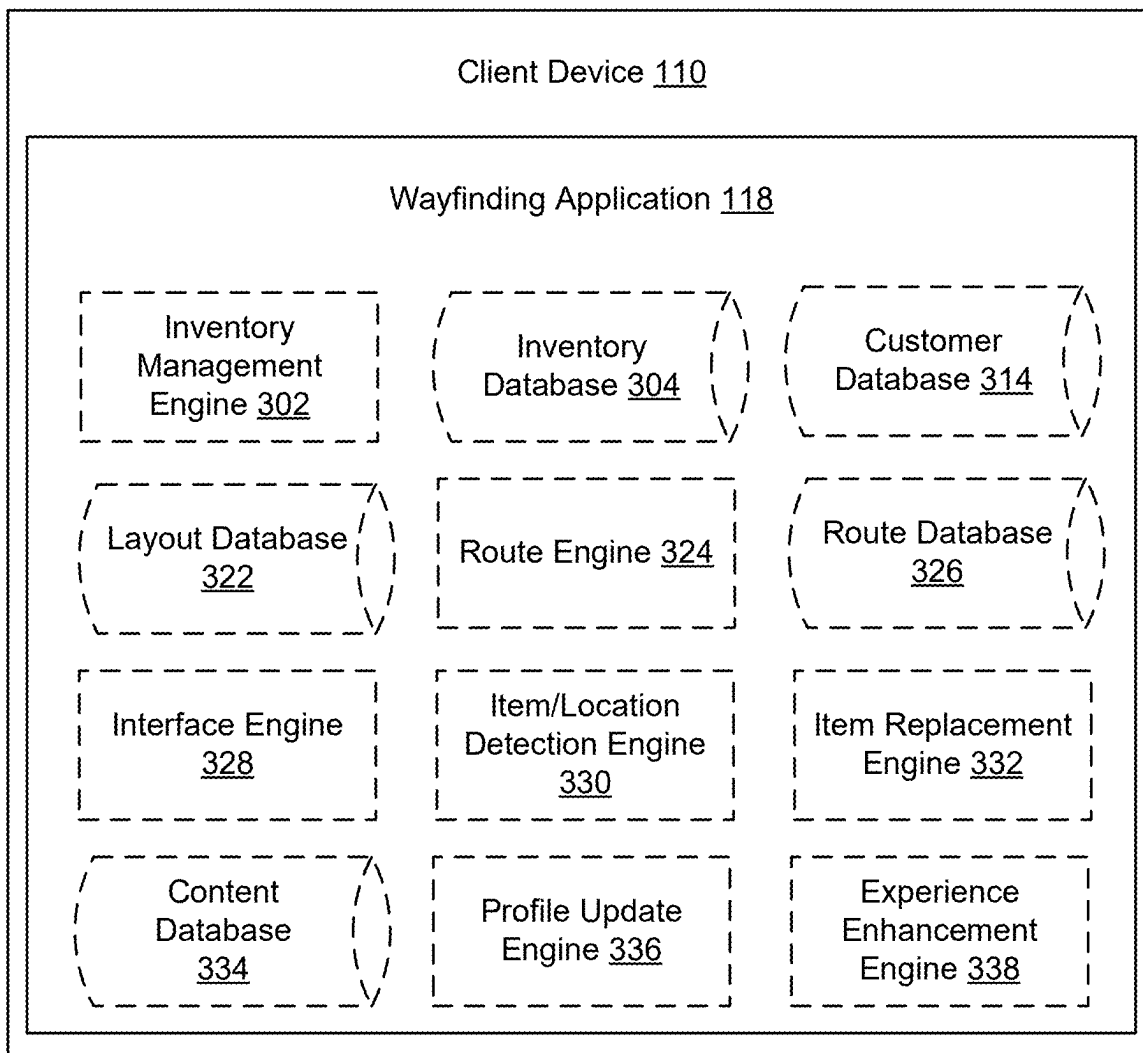


FIG. 3B

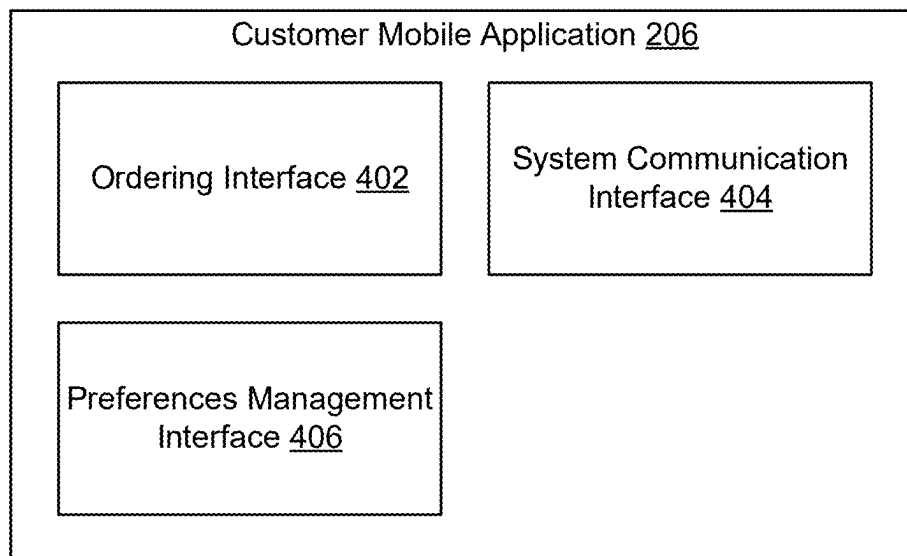


FIG. 4A

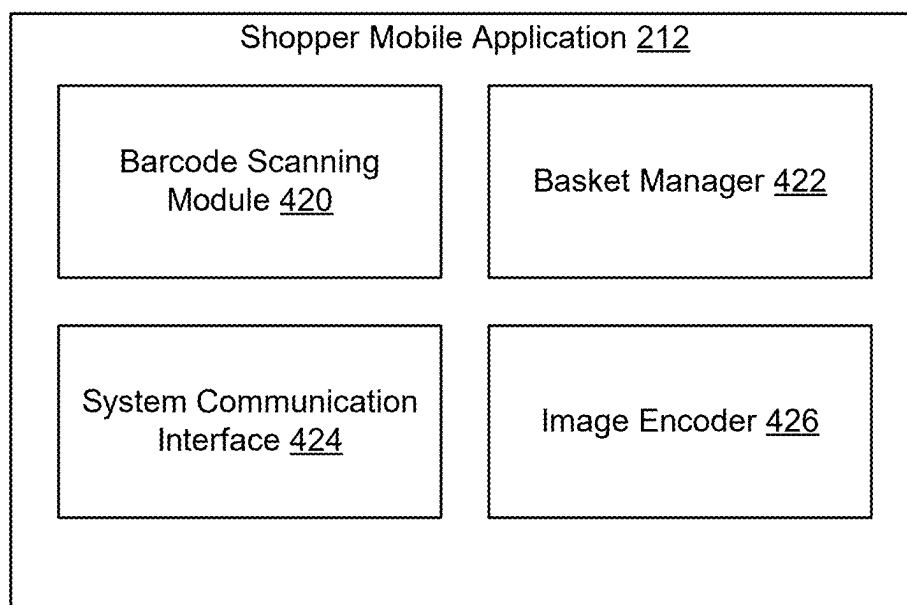


FIG. 4B

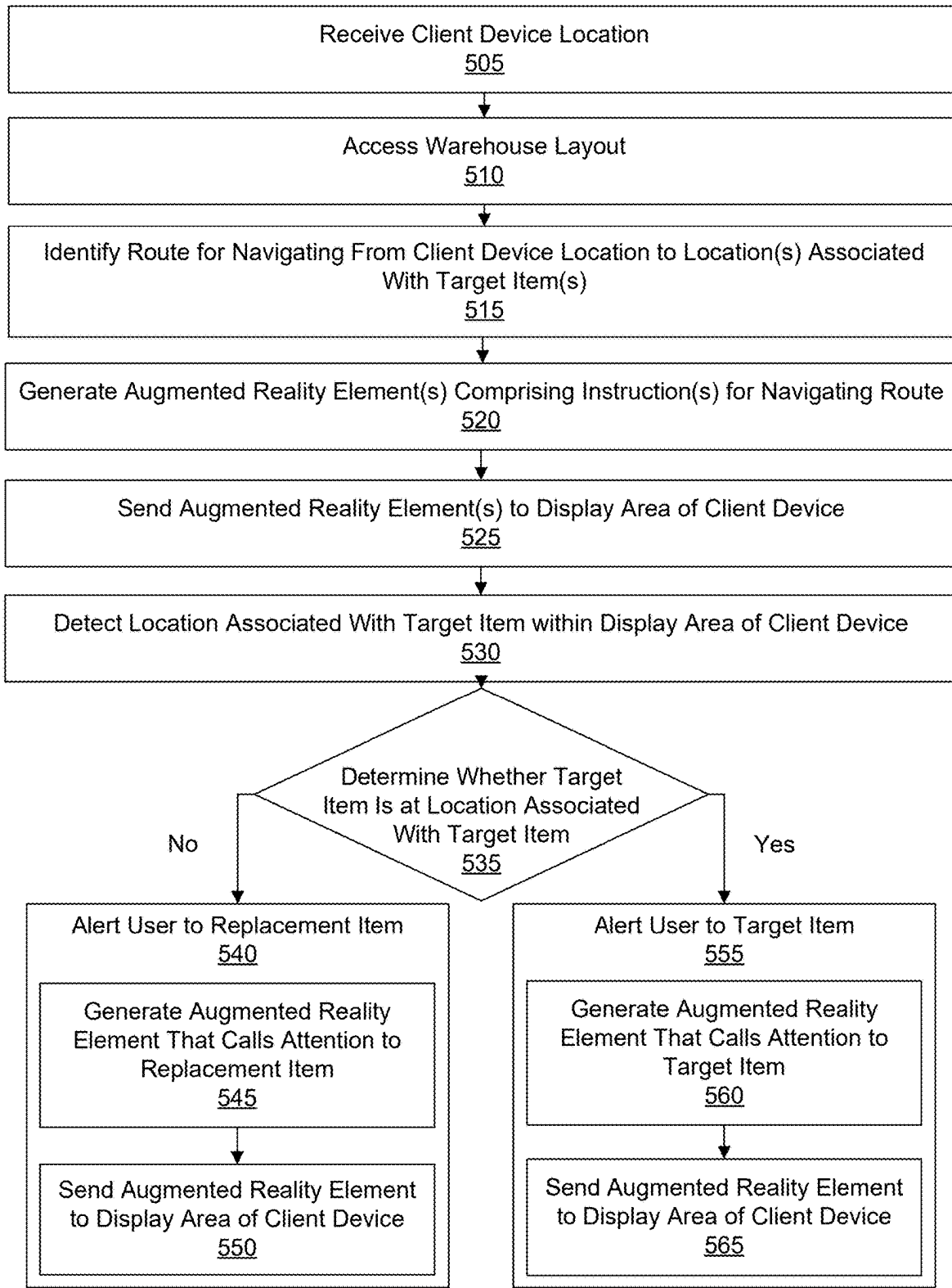


FIG. 5

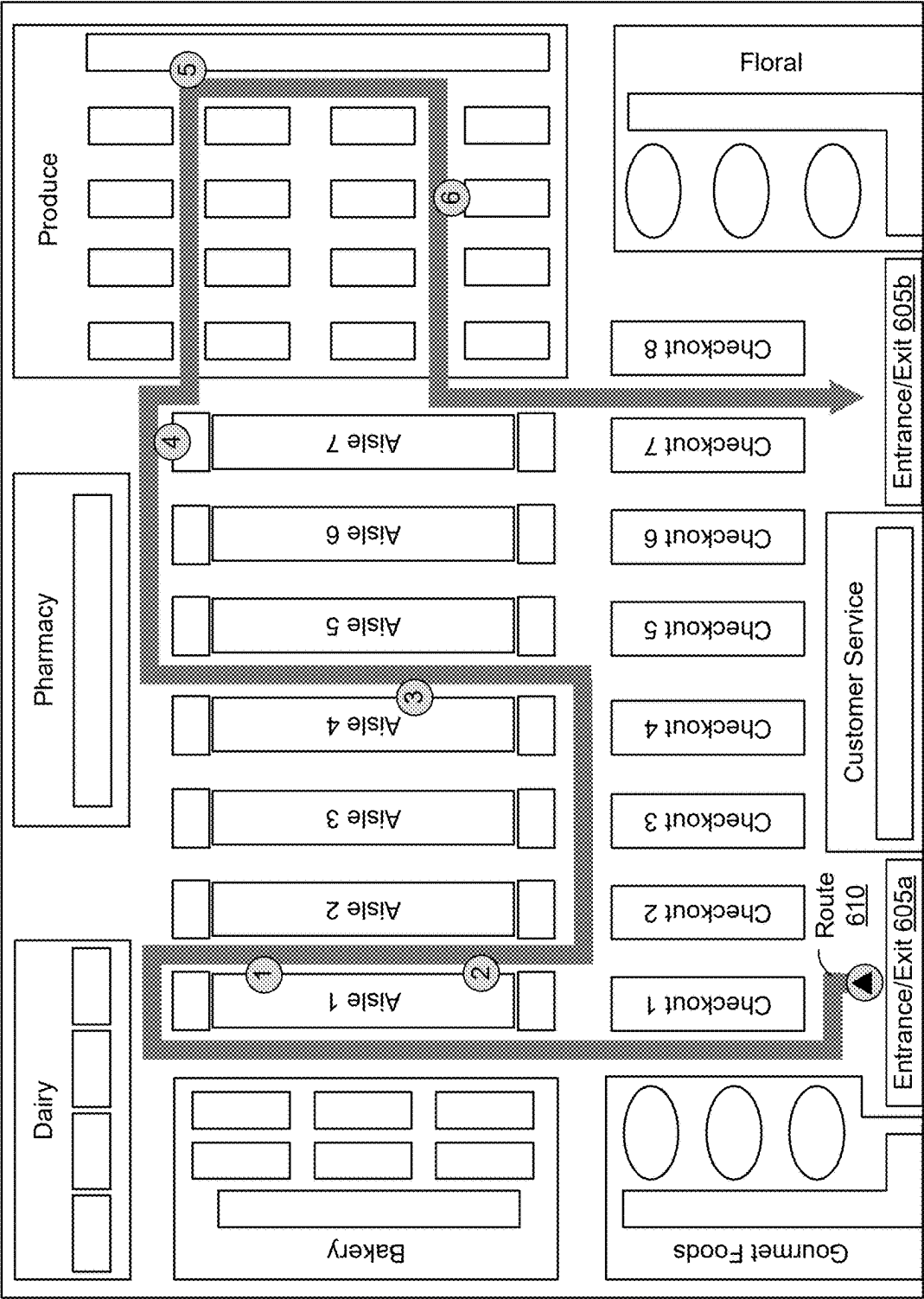


FIG. 6

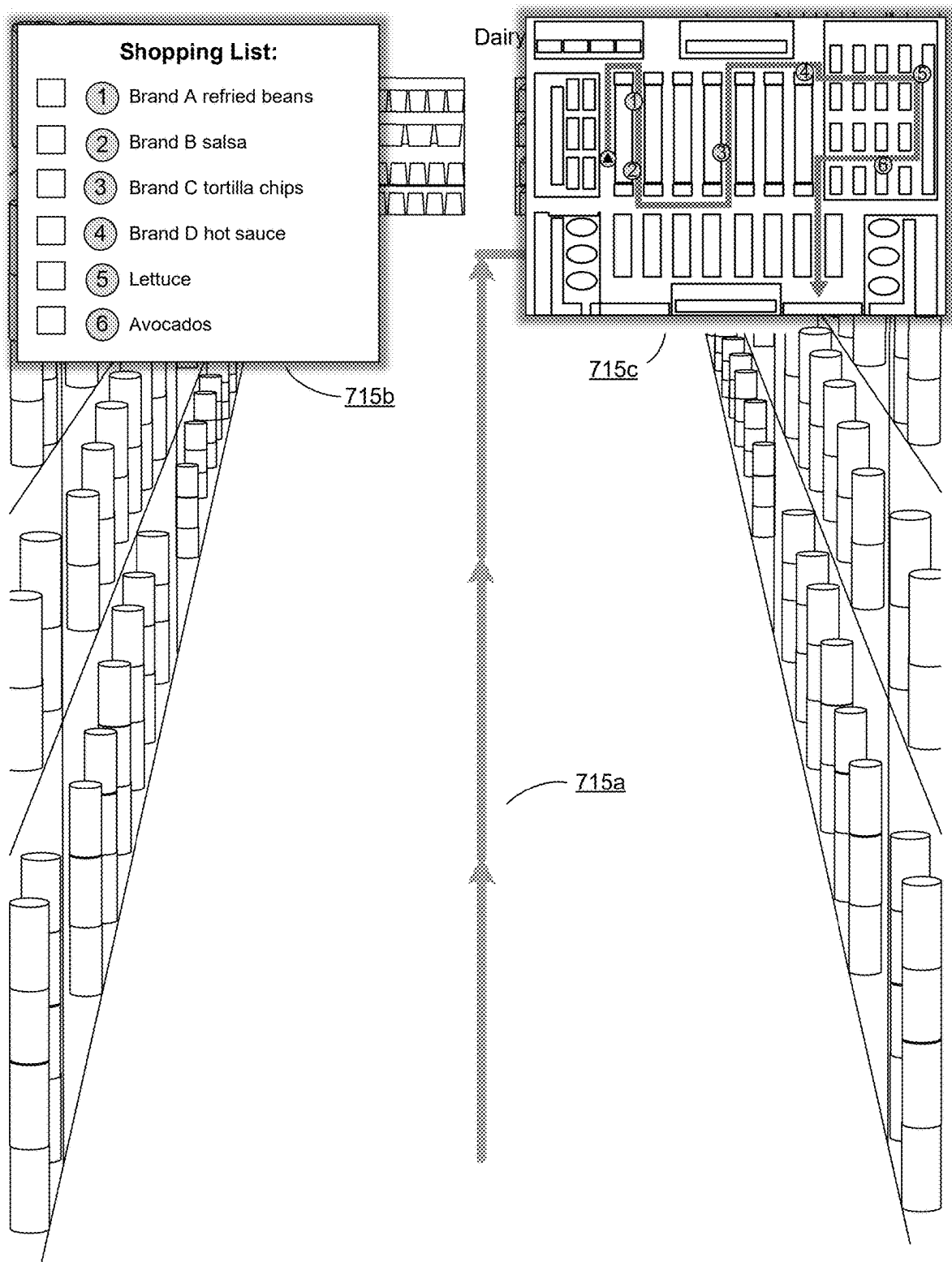


FIG. 7A

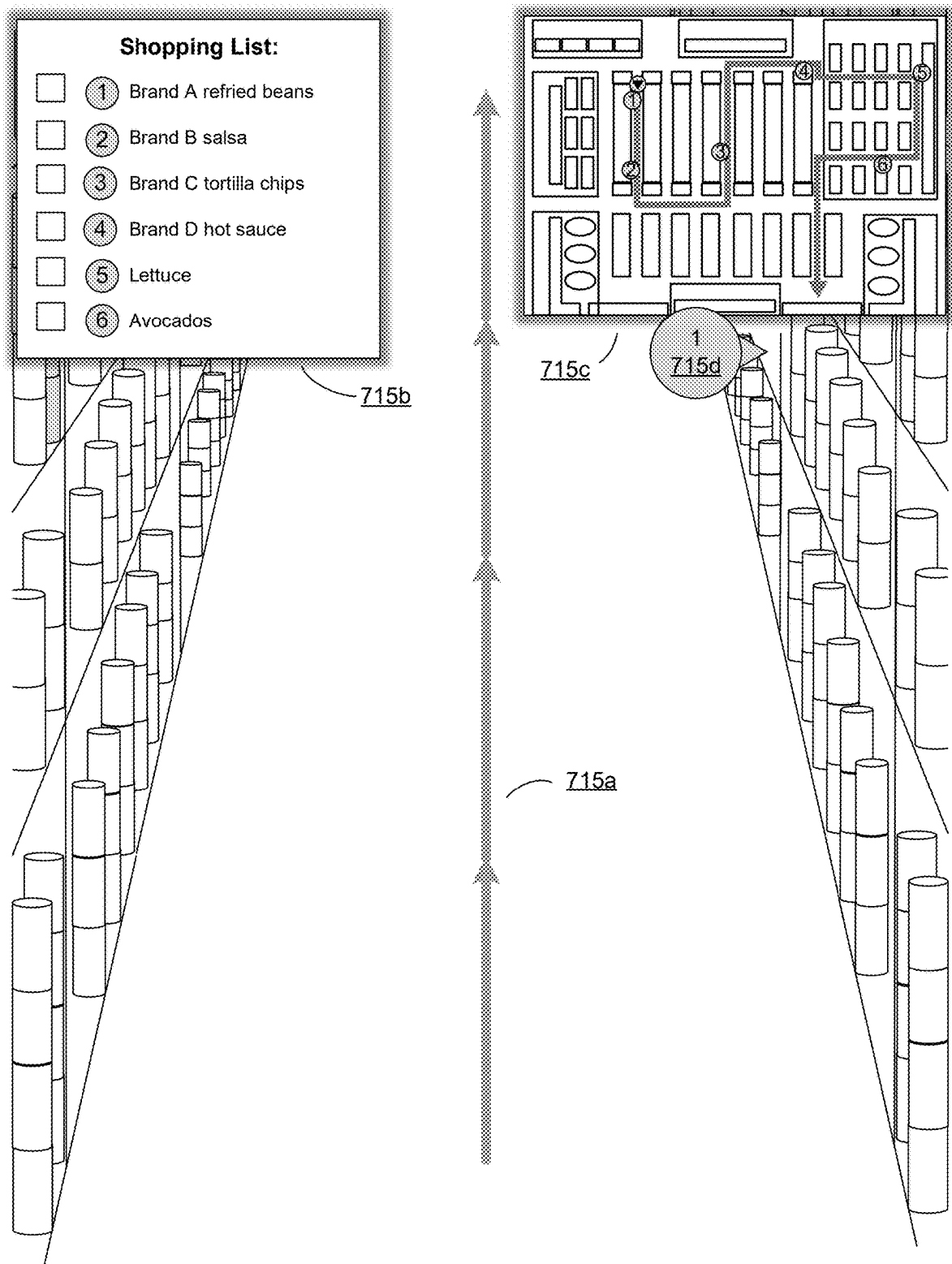


FIG. 7B

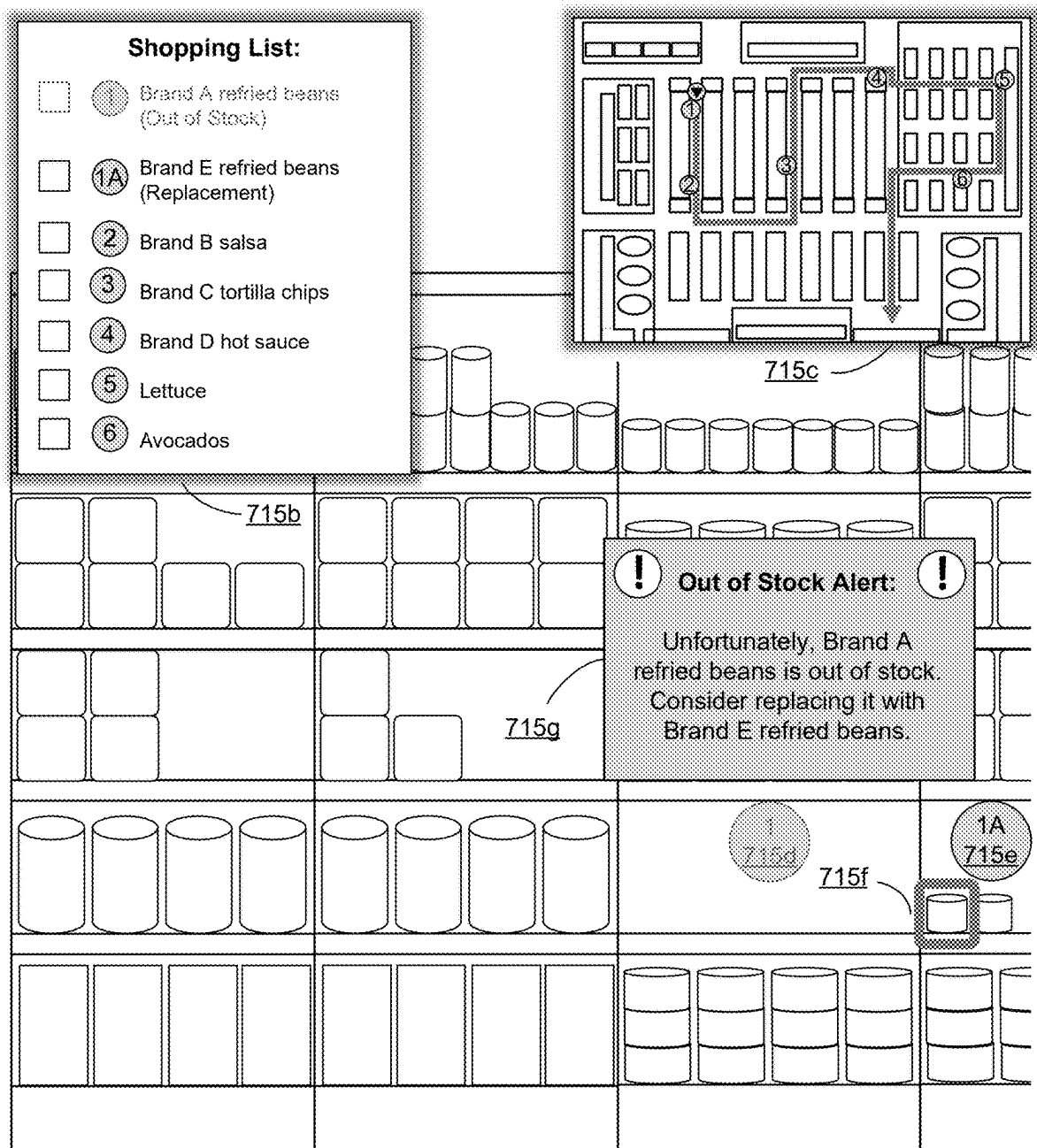


FIG. 7C

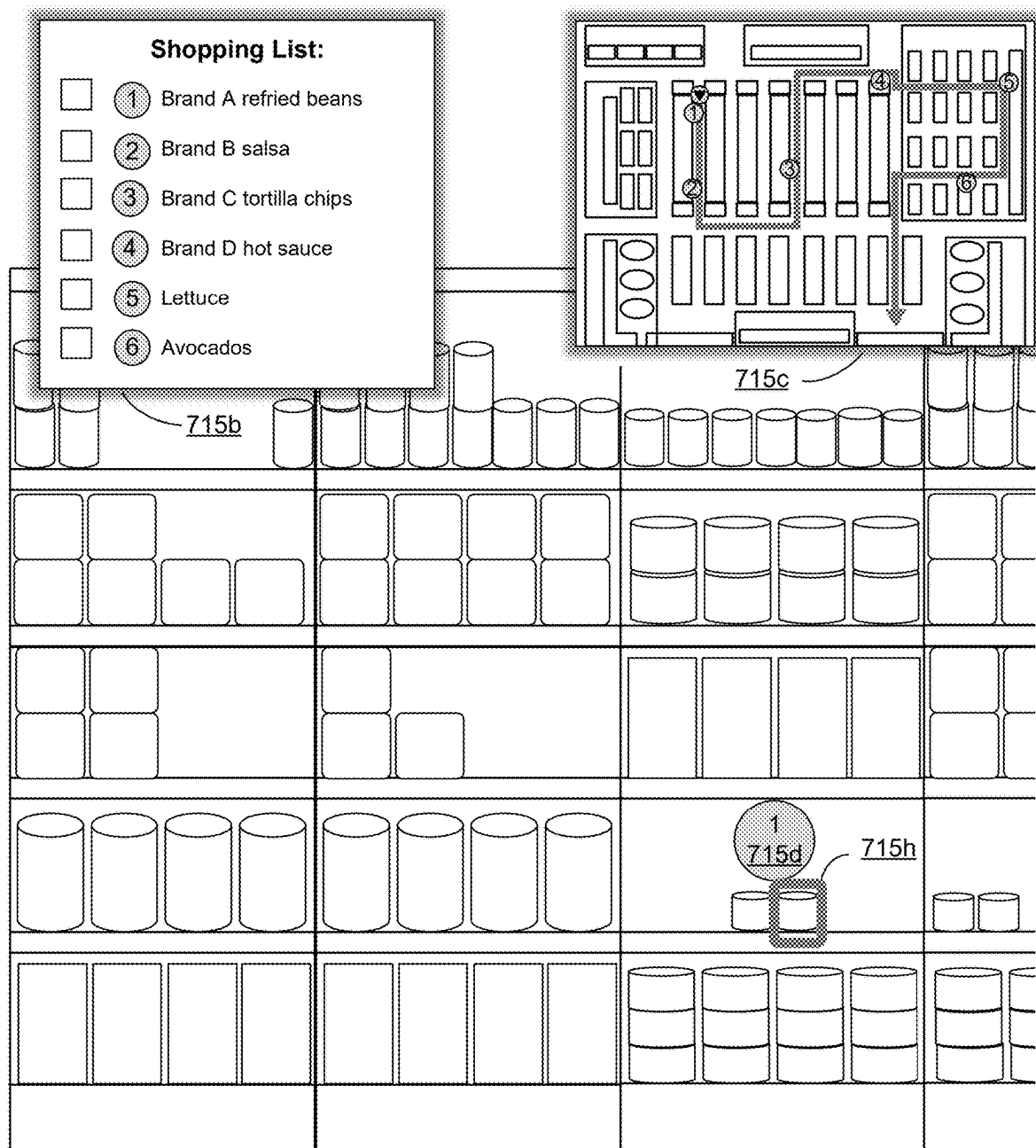


FIG. 7D

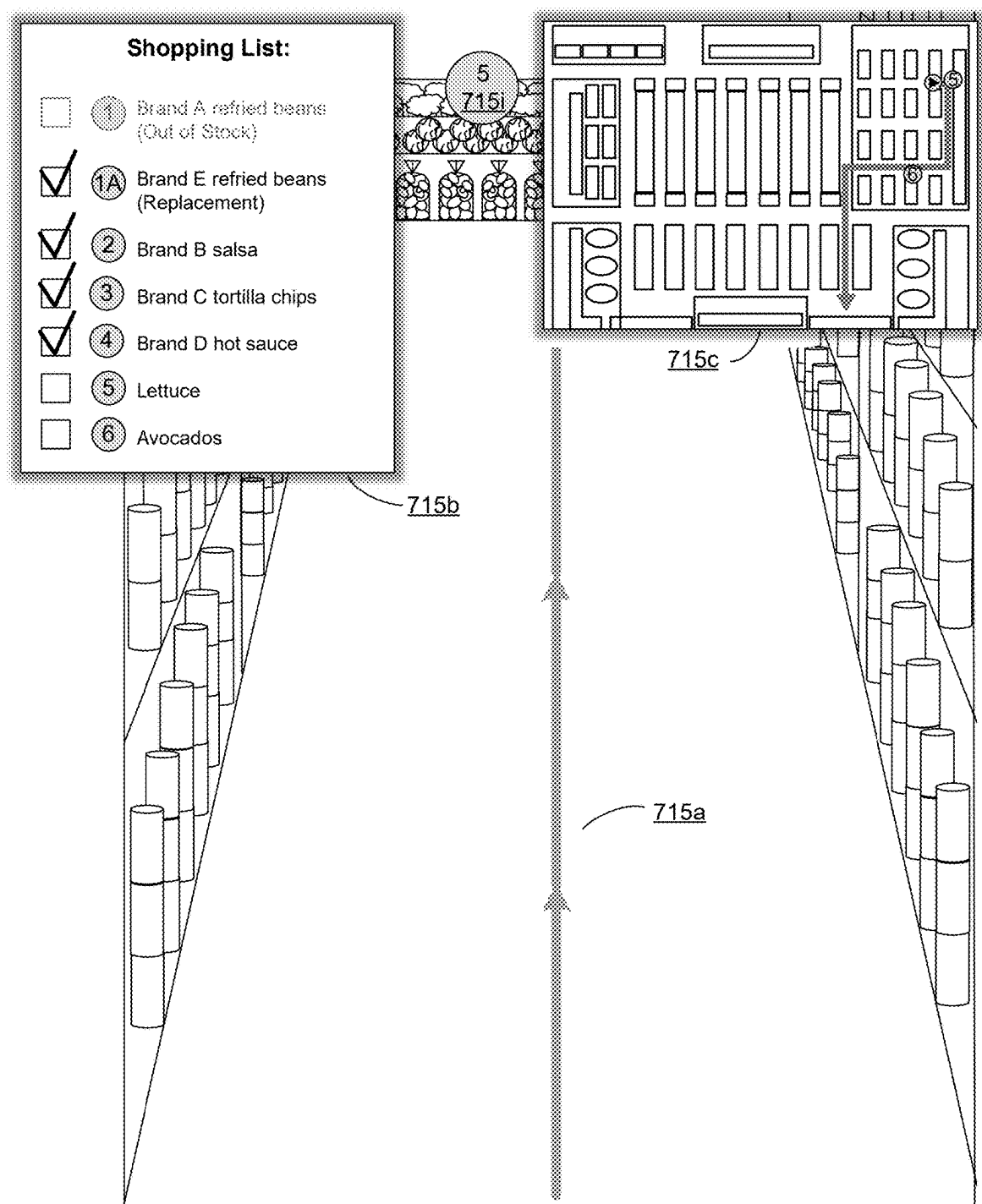


FIG. 7E

**METHOD, COMPUTER PROGRAM
PRODUCT, AND SYSTEM FOR DISPLAYING
AUGMENTED REALITY ELEMENTS FOR
NAVIGATING TO A LOCATION OF AN ITEM
WITHIN A WAREHOUSE**

**CROSS REFERENCE TO RELATED
APPLICATIONS**

[0001] This application is a continuation of co-pending U.S. application Ser. No. 18/236,370, filed Aug. 21, 2023, which is a continuation of U.S. application Ser. No. 17/877,718, filed Jul. 29, 2022, now U.S. Pat. No. 11,783,400, each of which is incorporated by reference in its entirety.

BACKGROUND

[0002] This disclosure relates generally to computer hardware and software for displaying augmented reality elements and more specifically to computer hardware and software for displaying augmented reality elements for navigating to a location of an item within a warehouse.

[0003] Warehouses, such as grocery stores, discount stores, department stores, and other physical retail stores, generally have layouts that group similar items included among their inventories together, making the items easier to find. For example, in a grocery store, bread, cupcakes, and donuts may be found in an aisle of a bakery department while pet food, leashes, and pet toys may be found in a different aisle for pet supplies. Over time, customers who frequent the same warehouses become familiar with their layouts, allowing them to navigate the warehouses more efficiently when shopping. For example, a customer who has shopped at the same warehouse for several years is likely to be familiar with the layout of the warehouse and may plan an efficient route for collecting items in their shopping list based on their knowledge of the layout, allowing them to navigate directly to the departments and aisles at which the items are located.

[0004] However, since the layouts of different warehouses may vary even if the warehouses belong to the same retail chain, customers, as well as shoppers who may be fulfilling orders on behalf of customers using online concierge systems, may not be familiar with the layouts of the warehouses at which they are shopping. For example, a customer or a shopper who is shopping at a grocery store for the first time may visit nearly every department and aisle of the store to search for a few items in a shopping list. In this example, the customer or shopper may need to return to some locations multiple times during the shopping trip and may be unable to find some of the items if they are not looking for them in the correct locations.

[0005] Furthermore, even customers or shoppers who are familiar with a layout of a warehouse may have trouble finding replacements for items that are out of stock. Moreover, finding replacements for items may be especially difficult if potential replacements are not located near the items they are replacing. For example, if a grocery store is out of unshelled almonds in a baking aisle, a customer or shopper who has only purchased unshelled almonds in the past may have trouble finding a replacement if they are not aware of potential replacements or their locations within the grocery store (e.g., whole shelled almonds in a snack aisle or slivered almonds in a different aisle near salad dressings).

SUMMARY

[0006] Warehouses generally have layouts that group similar items included among their inventories together, making the items easier to find. Over time, customers who frequent the same warehouses become familiar with their layouts, allowing the customers to navigate the warehouses more efficiently when shopping. However, since the layouts of different warehouses may vary, customers, as well as shoppers who may be fulfilling orders on behalf of customers using online concierge systems, may not be familiar with the layouts of the warehouses at which they are shopping. Additionally, customers or shoppers who are familiar with a layout of a warehouse may still have trouble finding replacements for items that are out of stock, especially if potential replacements are not located near the items they are replacing.

[0007] To help customers and shoppers navigate warehouses more efficiently and find replacements for items that are out of stock, a wayfinding application executing on a client device displays augmented reality elements for navigating to a location of an item within a warehouse, in accordance with one or more aspects of the disclosure. More specifically, the wayfinding application receives a current location of the client device within the warehouse and accesses a layout of the warehouse describing locations of items included among an inventory of the warehouse. The wayfinding application identifies a route from the current location of the client device to one or more locations within the warehouse associated with one or more target items, generates augmented reality elements including instructions for navigating the route, and sends the augmented reality elements to a display area of the client device. The wayfinding application then detects a location within the warehouse associated with a target item and determines whether the target item is at the location based on an image captured by the client device. Upon determining the target item is not at the location, the wayfinding application alerts a user of the client device to a replacement item by generating an augmented reality element that calls attention to the replacement item and sending the element to the display area of the client device.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a block diagram of a system environment in which an online system, such as an online concierge system, and an application, such as a wayfinding application, operate, according to one or more embodiments.

[0009] FIG. 2 illustrates an environment of an online concierge system and a wayfinding application, according to one or more embodiments.

[0010] FIG. 3A is a diagram of an online concierge system, according to one or more embodiments.

[0011] FIG. 3B is a diagram of a wayfinding application, according to one or more embodiments.

[0012] FIG. 4A is a diagram of a customer mobile application (CMA), according to one or more embodiments.

[0013] FIG. 4B is a diagram of a shopper mobile application (SMA), according to one or more embodiments.

[0014] FIG. 5 is a flowchart of a method for displaying augmented reality elements for navigating to a location of an item within a warehouse, according to one or more embodiments.

[0015] FIG. 6 illustrates a route for navigating from a location of a client device within a warehouse to one or more locations within the warehouse associated with one or more items, according to one or more embodiments.

[0016] FIG. 7A illustrates a set of augmented reality elements for navigating a route within a warehouse to one or more locations associated with one or more items, according to one or more embodiments.

[0017] FIG. 7B illustrates a set of augmented reality elements for navigating a route within a warehouse to one or more locations associated with one or more items, according to one or more embodiments.

[0018] FIG. 7C illustrates a set of augmented reality elements that calls attention to a replacement item within a warehouse, according to one or more embodiments.

[0019] FIG. 7D illustrates a set of augmented reality elements that calls attention to an item within a warehouse, according to one or more embodiments.

[0020] FIG. 7E illustrates a set of augmented reality elements for navigating a route within a warehouse to one or more locations associated with one or more items, according to one or more embodiments.

[0021] The figures depict embodiments of the present disclosure for purposes of illustration only. Alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles, or benefits touted, of the disclosure described herein.

DETAILED DESCRIPTION

System Architecture

[0022] FIG. 1 is a block diagram of a system environment 100 in which an online system, such as an online concierge system 102 as further described below in conjunction with FIGS. 2 and 3A, and an application, such as a wayfinding application 118 as further described below in conjunction with FIGS. 2 and 3B, operate. The system environment 100 shown in FIG. 1 comprises one or more client devices 110, a network 120, one or more third-party systems 130, the online concierge system 102, and one or more warehouses 150. In alternative configurations, different and/or additional components may be included in the system environment 100. Additionally, in other embodiments, the online concierge system 102 may be replaced by an online system configured to retrieve content for display to users and to transmit the content to one or more client devices 110 for display.

[0023] The client devices 110 are one or more computing devices capable of receiving user input as well as transmitting and/or receiving data via the network 120. In one embodiment, a client device 110 is a computer system, such as a desktop or a laptop computer. Alternatively, a client device 110 may be a device having computer functionality, such as a personal digital assistant (PDA), a mobile telephone, a smartphone, an augmented reality device, or another suitable device. A client device 110 is configured to communicate via the network 120. In one embodiment, a client device 110 executes an application allowing a user of the client device 110 to interact with the online concierge system 102. For example, the client device 110 executes a customer mobile application 206 or a shopper mobile application 212, as further described below in conjunction with FIGS. 4A and 4B, respectively, to enable interaction between the client device 110 and the online concierge

system 102. As an additional example, a client device 110 executes a browser application to enable interaction between the client device 110 and the online concierge system 102 via the network 120. In another embodiment, a client device 110 interacts with the online concierge system 102 through an application programming interface (API) running on a native operating system of the client device 110, such as IOS® or ANDROID™.

[0024] A client device 110 includes one or more processors 112 configured to control operation of the client device 110 by performing various functions. In various embodiments, a client device 110 includes a memory 114 comprising a non-transitory storage medium on which instructions are encoded. The memory 114 may have instructions encoded thereon that, when executed by the processor 112, cause the processor 112 to perform functions to execute the customer mobile application 206 or the shopper mobile application 212 to provide the functions further described below in conjunction with FIGS. 4A and 4B, respectively. In some embodiments, the memory 114 also or alternatively may have instructions encoded thereon corresponding to the modules further described below in conjunction with FIG. 3B that, when executed by the processor 112, cause the processor 112 to perform the functionality further described below in conjunction with FIGS. 2 and 5-7E. For example, the memory 114 has instructions encoded thereon that, when executed by the processor 112, cause the processor 112 to display augmented reality elements for navigating to a location of an item within a warehouse 150. In various embodiments, the wayfinding application 118 may be configured to access information associated with a client device 110 and/or components of the client device 110 (e.g., a camera, a display area, etc.). For example, the wayfinding application 118 may access location information (e.g., GPS coordinates) associated with a client device 110 and image or video data captured by a camera of the client device 110. In some embodiments, the wayfinding application 118 may be replaced by an application configured to retrieve content for display to users and to transmit the content to one or more client devices 110 for display.

[0025] The client devices 110 are configured to communicate via the network 120, which may comprise any combination of local area and/or wide area networks, using both wired and/or wireless communication systems. In one embodiment, the network 120 uses standard communications technologies and/or protocols. For example, the network 120 includes communication links using technologies such as Ethernet, 802.11, worldwide interoperability for microwave access (WiMAX), 3G, 4G, 5G, code division multiple access (CDMA), digital subscriber line (DSL), etc. Examples of networking protocols used for communicating via the network 120 include multiprotocol label switching (MPLS), transmission control protocol/Internet protocol (TCP/IP), hypertext transport protocol (HTTP), simple mail transfer protocol (SMTP), and file transfer protocol (FTP). Data exchanged over the network 120 may be represented using any suitable format, such as hypertext markup language (HTML) or extensible markup language (XML). In some embodiments, all or some of the communication links of the network 120 may be encrypted using any suitable technique or techniques.

[0026] One or more third-party systems 130 may be coupled to the network 120 for communicating with the online concierge system 102 or with the client device(s) 110.

In one embodiment, a third-party system **130** is an application provider communicating information describing applications for execution by a client device **110** or communicating data to client devices **110** for use by an application executing on the client device **110**. In other embodiments, a third-party system **130** provides content or other information for presentation via a client device **110**. For example, a third-party system **130** stores one or more web pages and transmits the web pages to a client device **110** or to the online concierge system **102**. A third-party system **130** may also communicate information to the online concierge system **102**, such as advertisements, content, or information about an application provided by the third-party system **130**.

[0027] The online concierge system **102** includes one or more processors **142** configured to control operation of the online concierge system **102** by performing various functions. In various embodiments, the online concierge system **102** includes a memory **144** comprising a non-transitory storage medium on which instructions are encoded. The memory **144** may have instructions encoded thereon corresponding to the modules further described below in conjunction with FIG. 3A that, when executed by the processor **142**, cause the processor **142** to perform the functionality further described below in conjunction with FIGS. 2 and 5-7E. For example, the memory **144** has instructions encoded thereon that, when executed by the processor **142**, cause the processor **142** to display augmented reality elements for navigating to a location of an item within a warehouse **150**. Additionally, the online concierge system **102** includes a communication interface configured to connect the online concierge system **102** to one or more networks, such as network **120**, or to otherwise communicate with devices (e.g., client devices **110**) connected to the network(s). In various embodiments, the online concierge system **102** may be configured to access information associated with a client device **110** and/or components of the client device **110** (e.g., a camera, a display area, etc.). For example, the online concierge system **102** may access location information (e.g., GPS coordinates) associated with a client device **110** and image or video data captured by a camera of the client device **110**.

[0028] One or more warehouses **150** may be coupled to the network **120** for communicating various types of information with the online concierge system **102**, the client device(s) **110**, and/or the third-party system(s) **130**. In some embodiments, a warehouse **150** (e.g., a physical grocery store) includes a location tracking system **152** capable of transmitting data to and/or receiving data from the wayfinding application **118** and/or the online concierge system **102** via the network **120**. In such embodiments, the location tracking system **152** may include a surveillance system **154** and/or one or more sensors **156** that track(s) the locations of client devices **110** within the warehouse **150**. For example, a location tracking system **152** at a warehouse **150** may include one or more sensors **156** that detect client devices **110** scanning for Wi-Fi networks and record MAC addresses associated with any detected client devices **110**. In this example, the location tracking system **152** may track the location of a client device **110** (e.g., in various aisles or departments) as it moves throughout the warehouse **150** based on its MAC address and image or video data of a user of the client device **110** captured by one or more cameras included in a surveillance system **154** of the location tracking system **152**. Continuing with this example, the location

tracking system **152** may transmit data to the wayfinding application **118** and/or the online concierge system **102** describing the location of the client device **110** within the warehouse **150**. A location tracking system **152** may transmit data describing a location of a client device **110** within a warehouse **150** to the wayfinding application **118** and/or the online concierge system **102** in association with various types of information (e.g., information identifying the warehouse **150**, a time at which the client device **110** was at the location, etc.). In the above example, the location tracking system **152** may transmit the data describing the location of the client device **110** within the warehouse **150** to the wayfinding application **118** and/or the online concierge system **102** in association with a warehouse identifier associated with the warehouse **150** and a timestamp indicating a date and time at which the client device **110** was at the location within the warehouse **150**.

[0029] One or more of a client device **110**, a third-party system **130**, a location tracking system **152**, or the online concierge system **102** may be special-purpose computing devices configured to perform specific functions, as further described below in conjunction with FIGS. 2-7E, and may include specific computing components such as processors, memories, communication interfaces, and/or the like.

System Overview

[0030] FIG. 2 illustrates an environment **200** of an online platform, such as an online concierge system **102**, and an application, such as a wayfinding application **118**, according to one or more embodiments. The figures use like reference numerals to identify like elements. A letter after a reference numeral, such as “**150a**,” indicates that the text refers specifically to the element having that particular reference numeral. A reference numeral in the text that is not followed by a letter, such as “**150**,” refers to any or all of the elements in the figures bearing that reference numeral. For example, “**150**” in the text may refer to reference numerals “**150a**,” “**150b**,” and/or “**150c**” in the figures.

[0031] The environment **200** includes an online concierge system **102**. The online concierge system **102** is configured to receive orders from one or more customers **204** (only one is shown for the sake of simplicity). An order specifies a list of goods (items or products) to be delivered to a customer **204**. An order also specifies a location to which goods are to be delivered, and a time window during which the goods should be delivered. In some embodiments, an order specifies one or more retailers from which goods should be purchased. A customer **204** may use a customer mobile application (CMA) **206**, which is configured to communicate with the online concierge system **102**, to place an order.

[0032] The online concierge system **102** is configured to transmit orders received from customers **204** to one or more shoppers **208**. A shopper **208** may be a person (e.g., a contractor, an employee, etc.), an entity, or an autonomous device (e.g., a robot) enabled to fulfill orders received by the online concierge system **102**. A shopper **208** travels between a warehouse **150** and a delivery location (e.g., a customer's home or office) and may do so by car, truck, bicycle, scooter, foot, or via any other mode of transportation. In some embodiments, a delivery may be partially or fully automated, e.g., using a self-driving car. The environment **200** also includes three warehouses **150a**, **150b**, and **150c** (while only three are shown for the sake of simplicity, the environment **200** may include hundreds of warehouses **150**). The

warehouses **150** may be physical retailers, such as grocery stores, discount stores, department stores, etc., or non-public warehouses **150** storing items that may be collected and delivered to customers **204**. Each shopper **208** fulfills an order received from the online concierge system **102** at one or more warehouses **150**, delivers the order to a customer **204**, or performs both fulfillment and delivery. In one embodiment, shoppers **208** make use of a shopper mobile application **212** which is configured to interact with the online concierge system **102**.

[0033] As described above, the wayfinding application **118** is configured to display augmented reality elements for navigating to a location of an item within a warehouse **150**. The wayfinding application **118** may be executed on a client device **110** associated with a shopper **208** fulfilling an order on behalf of a customer **204**. Alternatively, the wayfinding application **118** may be executed on a client device **110** associated with a customer **204** of a retailer associated with a warehouse **150**, in which the customer **204** may or may not be associated with the online concierge system **102**. For example, the wayfinding application **118** may be executed on a client device **110** associated with a customer **204** of a retailer associated with a warehouse **150**, in which the customer **204** is doing their own shopping at the warehouse **150** and may or may not be associated with the online concierge system **102**. In various embodiments, the wayfinding application **118** may interact with the online concierge system **102**, the CMA **206**, and/or the SMA **212**, while in other embodiments, the wayfinding application **118** is a standalone application. In some embodiments, the online concierge system **102** is configured to perform some or all of the functionality described below in conjunction with FIG. 3A, while in other embodiments, the wayfinding application **118** is configured to perform some or all of the functionality described below in conjunction with FIG. 3B.

[0034] FIG. 3A is a diagram of an online concierge system **102**, according to one or more embodiments. In various embodiments, the online concierge system **102** may include different or additional modules than those described in conjunction with FIG. 3A. Furthermore, in some embodiments, the online concierge system **102** includes fewer modules than those described in conjunction with FIG. 3A.

[0035] The online concierge system **102** includes an inventory management engine **302**, which interacts with inventory systems associated with each warehouse **150**. In one embodiment, the inventory management engine **302** requests and receives inventory information maintained by a warehouse **150**. The inventory of each warehouse **150** is unique and may change over time. The inventory management engine **302** monitors changes in inventory for each participating warehouse **150**. The inventory management engine **302** is also configured to store inventory records in an inventory database **304**. The inventory database **304** may store information in separate records—one for each participating warehouse **150**—or may consolidate or combine inventory information into a unified record. Inventory information includes attributes of items that include both qualitative and quantitative information about the items, including size, color, weight, SKU, serial number, etc. In some embodiments, the inventory management engine **302** may update the inventory of a warehouse **150** based on information received from the item/location detection engine **330** (described below) describing a quantity of an item detected within the warehouse **150**. For example, if information

stored in the inventory database **304** indicates that an item is out of stock at a warehouse **150**, but the item/location detection engine **330** detects three of the item at the warehouse **150**, the inventory management engine **302** may update the inventory database **304** to reflect this. In one embodiment, the inventory database **304** also stores purchasing rules associated with each item, if they exist. For example, age-restricted items such as alcohol and tobacco are flagged accordingly in the inventory database **304**. Additional inventory information useful for predicting the availability of items may also be stored in the inventory database **304**. For example, for each item-warehouse combination (a particular item at a particular warehouse **150**), the inventory database **304** may store a time that the item was last found, a time that the item was last not found (e.g., if a shopper **208** looked for the item but could not find it), a rate at which the item is found, and a popularity of the item. In some embodiments, the inventory management engine **302** receives an item catalog from a warehouse **150** identifying items offered for purchase by the warehouse **150**. In such embodiments, one or more images of each item included in the item catalog may be stored in the inventory database **304** in association with various types of information (e.g., information identifying the item and the warehouse **150**, a date the catalog was received, etc.).

[0036] For each item, the inventory database **304** identifies one or more attributes of the item and corresponding values for each attribute of the item. For example, the inventory database **304** includes an entry for each item offered by a warehouse **150**, in which an entry for an item includes an item identifier that uniquely identifies the item. The entry includes different fields, with each field corresponding to an attribute of the item. A field of an entry includes a value for an attribute corresponding to the field, allowing the inventory database **304** to maintain values of different attributes for various items.

[0037] In various embodiments, the inventory management engine **302** maintains a taxonomy of items offered for purchase by one or more warehouses **150**. For example, the inventory management engine **302** receives an item catalog from a warehouse **150** identifying items offered for purchase by the warehouse **150**. From the item catalog, the inventory management engine **302** determines a taxonomy of items offered by the warehouse **150**, in which different levels of the taxonomy provide different levels of specificity about items included in the levels. In various embodiments, the taxonomy identifies a category and associates one or more specific items with the category. For example, a category identifies “milk,” and the taxonomy associates identifiers of different milk items (e.g., milk offered by different brands, milk having one or more different attributes, etc.) with the category. Thus, the taxonomy maintains associations between a category and specific items offered by the warehouse **150** matching the category. In some embodiments, different levels of the taxonomy identify items with differing levels of specificity based on any suitable attribute or combination of attributes of the items. For example, different levels of the taxonomy specify different combinations of attributes of items, so items in lower levels of the hierarchical taxonomy have a greater number of attributes, corresponding to greater specificity in a category, while items in higher levels of the hierarchical taxonomy have a fewer number of attributes, corresponding to less specificity in a category. In various embodiments, higher levels of the

taxonomy include fewer details about items, so greater numbers of items are included in higher levels (e.g., higher levels include a greater number of items satisfying a broader category). Similarly, lower levels of the taxonomy include greater details about items, so fewer numbers of items are included in the lower levels (e.g., lower levels include a fewer number of items satisfying a more specific category). The taxonomy may be received from a warehouse 150 in various embodiments. In other embodiments, the inventory management engine 302 applies a trained classification model to an item catalog received from a warehouse 150 to include different items in levels of the taxonomy, so application of the trained classification model associates specific items with categories corresponding to levels within the taxonomy.

[0038] Inventory information provided by the inventory management engine 302 may supplement training datasets 320. Inventory information provided by the inventory management engine 302 may not necessarily include information about the outcome of fulfilling a delivery order associated with the item, whereas the data within the training datasets 320 is structured to include an outcome of fulfilling a delivery order (e.g., if an item in an order was or was not collected).

[0039] The online concierge system 102 also includes an order fulfillment engine 306 which is configured to synthesize and display an ordering interface to each customer 204 (for example, via the customer mobile application 206). The order fulfillment engine 306 is also configured to access the inventory database 304 in order to determine which items are available at which warehouse 150. The order fulfillment engine 306 may supplement the item availability information from the inventory database 304 with item availability information predicted by a machine-learned item availability model 316. The order fulfillment engine 306 determines a sale price for each item ordered by a customer 204. Prices set by the order fulfillment engine 306 may or may not be identical to in-store prices determined by retailers (which is the price that customers 204 and shoppers 208 would pay at the retail warehouses 150). The order fulfillment engine 306 also facilitates transactions associated with each order. In one embodiment, the order fulfillment engine 306 charges a payment instrument associated with a customer 204 when he/she places an order. The order fulfillment engine 306 may transmit payment information to an external payment gateway or payment processor. The order fulfillment engine 306 stores payment and transactional information associated with each order in a transaction records database 308.

[0040] In various embodiments, the order fulfillment engine 306 generates and transmits a search interface to a client device 110 of a customer 204 for display via the customer mobile application 206. The order fulfillment engine 306 receives a query comprising one or more terms from a customer 204 and retrieves items satisfying the query, such as items having descriptive information matching at least a portion of the query. In various embodiments, the order fulfillment engine 306 leverages item embeddings for items to retrieve items based on a received query. For example, the order fulfillment engine 306 generates an embedding for a query and determines measures of similarity between the embedding for the query and item embeddings for various items included in the inventory database 304.

[0041] In some embodiments, the order fulfillment engine 306 also shares order details with warehouses 150. For example, after successful fulfillment of an order, the order fulfillment engine 306 may transmit a summary of the order to the appropriate warehouses 150. Details of an order may indicate the items purchased, a total value of the items, and in some cases, an identity of a shopper 208 and a customer 204 associated with the order. In one embodiment, the order fulfillment engine 306 pushes transaction and/or order details asynchronously to retailer systems. This may be accomplished via the use of webhooks, which enable programmatic or system-driven transmission of information between web applications. In another embodiment, retailer systems may be configured to periodically poll the order fulfillment engine 306, which provides details of all orders which have been processed since the last request.

[0042] The order fulfillment engine 306 may interact with a shopper management engine 310, which manages communication with and utilization of shoppers 208. In one embodiment, the shopper management engine 310 receives a new order from the order fulfillment engine 306. The shopper management engine 310 identifies the appropriate warehouse 150 to fulfill the order based on one or more parameters, such as a probability of item availability determined by the machine-learned item availability model 316, the contents of the order, the inventory of the warehouses 150, and the proximity of the warehouses 150 to the delivery location. The shopper management engine 310 then identifies one or more appropriate shoppers 208 to fulfill the order based on one or more parameters, such as the shoppers' proximity to the appropriate warehouse 150 (and/or to the customer 204), his/her familiarity level with that particular warehouse 150, etc. Additionally, the shopper management engine 310 accesses a shopper database 312 which stores information describing each shopper 208, such as his/her name, gender, rating, previous shopping history, etc.

[0043] As part of fulfilling an order, the order fulfillment engine 306 and/or shopper management engine 310 may access a customer database 314, which stores profile information associated with each customer 204. In some embodiments, the customer database 314 may store profile information associated with each customer 204 who is a user of the online concierge system 102 (e.g., a customer 204 who has placed an order to be fulfilled by a shopper 208). In various embodiments, the customer database 314 also or alternatively may store profile information associated with each customer 204 of one or more retailers associated with one or more warehouses 150, in which a customer 204 may or may not be associated with the online concierge system 102 (e.g., a customer 204 of a retailer associated with a warehouse 150 who is doing their own shopping at the warehouse 150). Profile information stored in the customer database 314 may include each customer's name, address, gender, preferences, favorite items, stored payment instruments, order history, etc. In some embodiments, profile information associated with a customer 204 may describe one or more preferences of the customer 204 for items having various attributes. Preferences for items may describe types, qualities, quantities, colors, sizes, brands, manufacturing locations, ingredients, packaging, or any other attributes of items. For example, the customer database 314 may store information describing a preference of a customer 204 for certain types of milk (e.g., oat and soy). As an additional example, the customer database 314 may store

information describing a preference of a customer **204** for a particular brand of coffee creamer and a minimum number of days between an expiration date and a current date for dairy items.

[0044] In some embodiments, preferences stored in the customer database **314** also may include a customer's dislike for items having various attributes and/or a set of restrictions associated with the customer **204**. For example, if a customer **204** is allergic to peanuts, preferences included among the customer's profile information may indicate that the customer **204** has a dietary restriction related to peanuts and is therefore unlikely to be interested in food items including peanuts as an ingredient. As an additional example, if a customer **204** previously reported a problem with a delivery order, in which organic blueberries requested by the customer **204** were unavailable and were replaced with non-organic blueberries and the customer **204** indicated that the replacement item was a poor replacement, preferences included among the customer's profile information may indicate that the customer **204** dislikes non-organic blueberries.

[0045] Profile information associated with a customer **204** stored in the customer database **314** may include the customer's order and/or purchase history and/or one or more lists of items (e.g., shopping lists) associated with the customer **204**. Information describing an order/purchase included in a customer's order/purchase history may include a date and time of the order/purchase, information describing items included in the order/purchase, a price of each item included in the order/purchase, a total cost of the order/purchase, a warehouse **150** at which the order/purchase was fulfilled/completed, coupons, discounts, or other promotions applied to the order/purchase, or any other information that may be associated with an order/purchase. In embodiments in which the customer database **314** includes one or more lists of items associated with a customer **204**, each list may include one or more items specified by the customer **204**. Each list of items associated with a customer **204** may be created by the customer **204** using an interface generated and transmitted by the interface engine **328**, as described below. Similar to each order/purchase included in an order/purchase history, information describing a list of items may include a date the list was created, information describing items included in the list, information identifying a warehouse **150** at which items on the list were collected, or any other suitable types of information that may be associated with a list of items.

[0046] In various embodiments, profile information stored in the customer database **314** also may include one or more goals associated with a customer **204**. A goal associated with a customer **204** may describe the customer's personal goals (e.g., health goals, nutritional goals, etc.). For example, profile information stored in the customer database **314** may describe goals of a customer **204** to lose weight, increase protein intake, and eat fewer processed foods. As an additional example, profile information associated with a customer **204** may describe a goal of saving money on groceries.

[0047] In some embodiments, profile information stored in the customer database **314** also may include information received from the interface engine **328** (described below) describing augmented reality elements sent to display areas of client devices **110** associated with customers **204**. In such embodiments, the customers **204** may be users of the client

devices **110** or may have placed orders fulfilled by shoppers **208** who were using the client devices **110** when the augmented reality elements were sent. Information describing an augmented reality element sent to a display area of a client device **110** associated with a customer **204** may describe a function of the augmented reality element (e.g., to call attention to an item, to promote an item, to compare items, etc.) and identify one or more items or other physical elements to which the augmented reality element was directed. The information also may identify one or more content items included in the augmented reality element (e.g., a coupon, a recipe, a video, etc.) and a date and time that the augmented reality element was sent to the display area of the client device **110**.

[0048] In various embodiments, the order fulfillment engine **306** determines whether to delay display of a received order to shoppers **208** for fulfillment by a time interval. In response to determining to delay display of the received order by a time interval, the order fulfillment engine **306** evaluates subsequent orders received during the time interval for inclusion in one or more batches that also include the received order. After the time interval, the order fulfillment engine **306** displays the order to one or more shoppers **208** via the shopper mobile application **212**; if the order fulfillment engine **306** generated one or more batches including the received order and one or more subsequent orders received during the time interval, the batch(es) is/are also displayed to one or more shoppers **208** via the shopper mobile application **212**.

Machine Learning Models

[0049] The online concierge system **102** further includes the machine-learned item availability model **316**, a modeling engine **318**, and the training datasets **320**. The modeling engine **318** uses the training datasets **320** to generate the machine-learned item availability model **316**. The machine-learned item availability model **316** may learn from the training datasets **320**, rather than follow only explicitly programmed instructions. The inventory management engine **302**, order fulfillment engine **306**, and/or shopper management engine **310** may use the machine-learned item availability model **316** to determine a probability that an item is available at a warehouse **150**. The machine-learned item availability model **316** may be used to predict item availability for items being displayed to or selected by a customer **204** or included in received delivery orders. A single machine-learned item availability model **316** is used to predict the availability of any number of items.

[0050] The machine-learned item availability model **316** may be configured to receive, as inputs, information about an item, a warehouse **150** for collecting the item, and a time for collecting the item. The machine-learned item availability model **316** may be adapted to receive any information that the modeling engine **318** identifies as an indicator of item availability. At a minimum, the machine-learned item availability model **316** receives information about an item-warehouse pair, such as an item in a delivery order and a warehouse **150** at which the order may be fulfilled. Items stored in the inventory database **304** may be identified by item identifiers. As described above, various characteristics, some of which are specific to a warehouse **150** (e.g., a time that an item was last found in the warehouse **150**, a time that the item was last not found in the warehouse **150**, a rate at which the item is found, a popularity of the item, etc.) may

be stored for each item in the inventory database 304. Similarly, each warehouse 150 may be identified by a warehouse identifier and stored in a warehouse database along with information about the warehouse 150. A particular item at a particular warehouse 150 may be identified using an item identifier and a warehouse identifier. In other embodiments, the item identifier refers to a particular item at a particular warehouse 150, so that the same item at two different warehouses 150 is associated with two different identifiers. For convenience, both of these options to identify an item at a warehouse 150 are referred to herein as an “item-warehouse pair.” Based on the identifier(s), the online concierge system 102 may extract information about the item and/or warehouse 150 from the inventory database 304 and/or warehouse database and provide this extracted information as inputs to the machine-learned item availability model 316.

[0051] The machine-learned item availability model 316 contains a set of functions generated by the modeling engine 318 from the training datasets 320 that relate an item, a warehouse 150, timing information, and/or any other relevant inputs, to a probability that the item is available at the warehouse 150. Thus, for a given item-warehouse pair, the machine-learned item availability model 316 outputs a probability that the item is available at the warehouse 150. The machine-learned item availability model 316 constructs a relationship between the item-warehouse pair, the timing information, and/or any other inputs and the probability of availability (also referred to as “availability”) that is generic enough to apply to any number of different item-warehouse pairs. In some embodiments, the probability output by the machine-learned item availability model 316 includes a confidence score. The confidence score may be the error or uncertainty score of the probability of availability and may be calculated using any standard statistical error measurement. In some embodiments, the confidence score is based in part on whether the item-warehouse pair availability prediction was accurate for previous delivery orders (e.g., if an item was predicted to be available at a warehouse 150 and was not found by a shopper 208 or was predicted to be unavailable but was found by the shopper 208). In various embodiments, the confidence score is based in part on the age of the data for the item (e.g., if availability information has been received within the past hour or the past day). The set of functions of the machine-learned item availability model 316 may be updated and adapted following retraining with new training datasets 320. The machine-learned item availability model 316 may be any machine learning model, such as a neural network, a boosted tree, a gradient-boosted tree, or a random forest model. In some embodiments, the machine-learned item availability model 316 is generated from the XGBoost algorithm. The probability of availability of an item generated by the machine-learned item availability model 316 may be used to determine instructions delivered to a customer 204 and/or shopper 208, as described in further detail below.

[0052] The training datasets 320 relate a variety of different factors to known item availabilities from the outcomes of previous delivery orders (e.g., if an item was previously found or previously unavailable). The training datasets 320 include items included in previous delivery orders, whether the items in the previous delivery orders were collected, warehouses 150 associated with the previous delivery orders, and a variety of characteristics associated with each

of the items, which may be obtained from the inventory database 304. Each piece of data in the training datasets 320 includes an outcome of a previous delivery order (e.g., whether an item was collected). Item characteristics may be determined by the machine-learned item availability model 316 to be statistically significant factors predictive of an item’s availability. Item characteristics that are predictors of availability may be different for different items. For example, an item type factor might be the best predictor of availability for dairy items, whereas a time of day may be the best predictive factor of availability for vegetables. For each item, the machine-learned item availability model 316 may weight these factors differently, in which the weights are a result of a “learning” or a training process on the training datasets 320. The training datasets 320 are very large datasets taken across a wide cross section of warehouses 150, shoppers 208, items, delivery orders, times, and item characteristics. The training datasets 320 are large enough to provide a mapping from an item in an order to a probability that the item is available at a warehouse 150. In addition to previous delivery orders, the training datasets 320 may be supplemented by inventory information provided by the inventory management engine 302. In some embodiments, the training datasets 320 are historical delivery order information used to train the machine-learned item availability model 316, whereas the inventory information stored in the inventory database 304 includes factors input into the machine-learned item availability model 316 to determine an item availability for an item in a newly received delivery order. In various embodiments, the modeling engine 318 may evaluate the training datasets 320 to compare a single item’s availability across multiple warehouses 150 to determine if an item is chronically unavailable, which may indicate that the item is no longer manufactured. The modeling engine 318 may query a warehouse 150 through the inventory management engine 302 for updated item information about any such items.

Machine Learning Factors

[0053] The training datasets 320 include times associated with previous delivery orders. In some embodiments, the training datasets 320 include a time of day at which each previous delivery order was placed. Item availability may be impacted by time of day since items that are otherwise regularly stocked by warehouses 150 may become unavailable during high-volume shopping times. In addition, item availability may be affected by restocking schedules. For example, if a warehouse 150 mainly restocks items at night, item availability at the warehouse 150 will tend to decrease over the course of the day. Additionally, or alternatively, the training datasets 320 include a day of the week that previous delivery orders were placed. The day of the week may impact item availability since warehouses 150 may have reduced item inventory on popular shopping days and restocking shipments may be received on particular days. In some embodiments, the training datasets 320 include a time interval since an item was previously collected for a previous delivery order. If an item has recently been collected at a warehouse 150, this may increase the probability that it is still available. If a long interval of time has elapsed since an item has been collected, this may indicate that the probability that the item is available for subsequent orders is low or uncertain. In some embodiments, the training datasets 320 include a time interval since an item in a previous delivery

order was not found. If a short interval of time has elapsed since an item was not found, this may indicate that there is a low probability that the item will be available for subsequent delivery orders. Conversely, if a long interval of time has elapsed since an item was not found, this may indicate that the item may have been restocked and will be available for subsequent delivery orders. In some embodiments, the training datasets 320 may also include a rate at which an item is typically found by a shopper 208 at a warehouse 150, a number of days since inventory information about the item was last received from the inventory management engine 302, a number of times the item was not found during a previous week, or any number of additional rate-related or time-related information. Relationships between this rate-related and/or time-related information and item availability are determined by the modeling engine 318, which trains a machine learning model with the training datasets 320, producing the machine-learned item availability model 316.

[0054] The training datasets 320 include item characteristics. In some embodiments, the item characteristics include a department associated with an item. For example, if an item is yogurt, it is associated with a dairy department. Examples of departments include bakery, beverage, non-food, pharmacy, produce, floral, deli, prepared foods, meat, seafood, dairy, or any other categorization of items used by a warehouse 150. A department associated with an item may affect item availability since different departments have different item turnover rates and inventory levels. In some embodiments, the item characteristics include an aisle of a warehouse 150 associated with an item. The aisle of the warehouse 150 may affect item availability since different aisles of a warehouse 150 may be restocked more frequently than others. Additionally, or alternatively, the item characteristics may include an item popularity score. The item popularity score for an item may be proportional to the number of delivery orders received that include the item. An alternative or additional item popularity score may be provided by a retailer through the inventory management engine 302. In some embodiments, the item characteristics include a product type associated with an item. For example, if an item is a particular brand of a product, the product type will be a generic description of the product type, such as “milk” or “eggs.” The product type may affect item availability since certain product types may have higher turnover and restocking rates than others or may have larger inventories in the warehouses 150. In some embodiments, the item characteristics may include a number of times a shopper 208 was instructed to keep looking for an item after he or she was initially unable to find the item, a total number of delivery orders received for an item, whether or not an item is organic, vegan, gluten free, or any other characteristics associated with an item. The relationships between item characteristics and item availability are determined by the modeling engine 318, which trains a machine learning model with the training datasets 320, producing the machine-learned item availability model 316.

[0055] The training datasets 320 may include additional item characteristics that affect item availability and may therefore be used to build the machine-learned item availability model 316 relating a delivery order including an item to the item’s predicted availability. The training datasets 320 may be periodically updated with recent delivery orders. The training datasets 320 may be updated with item availability information provided directly from shoppers 208. Following

updating of the training datasets 320, the modeling engine 318 may retrain a model with the updated training datasets 320 and produce a new machine-learned item availability model 316.

Navigation Components

[0056] In some embodiments, the online concierge system 102 further includes a layout database 322. The layout database 322 stores one or more layouts of one or more warehouses 150. In some embodiments, a layout of a warehouse 150 may be received from a retailer associated with the warehouse 150 and updated by the retailer or the item/location detection engine 330, as described below. A layout of a warehouse 150 may be stored in the layout database 322 in association with information that uniquely identifies the warehouse 150, such as a name of the warehouse 150, a geographic location of the warehouse 150, etc. For example, a layout of a warehouse 150 may be stored in association with GPS coordinates associated with the warehouse 150, a combination of a name of a retailer associated with the warehouse 150 and a city and state in which the warehouse 150 is located, etc.

[0057] A layout of a warehouse 150 may describe departments and/or physical elements within the warehouse 150, such as organizational elements, including aisles, shelves within the aisles, various display areas (e.g., frozen or refrigerated display cases, display tables, and sample and promotional displays), etc., as well as their arrangement within the warehouse 150. In addition to organizational elements, a layout of a warehouse 150 also may describe additional physical elements within the warehouse 150, such as items included among an inventory of the warehouse 150, service counters (e.g., for various departments and/or check-out counters), and various building elements (e.g. entrances/exits, floors, ceilings, walls, stairs, elevators, etc.), and their arrangement within the warehouse 150. A layout of a warehouse 150 also may include one or more images of departments and physical elements within the warehouse 150 captured from one or more angles, positions, etc. In some embodiments, a layout of a warehouse 150 also may describe dimensions associated with the warehouse 150 (e.g., dimensions associated with departments and/or organizational elements within the warehouse 150). For example, a layout of a warehouse 150 may describe lengths of various aisles, distances between the aisles, heights of shelves, etc. Furthermore, in various embodiments, a layout of a warehouse 150 may describe congestion/traffic within the warehouse 150. For example, a layout of a warehouse 150 may describe certain times or days of the week when particular service counters or aisles with sample displays are more congested than others.

[0058] Additionally, a layout of a warehouse 150 may describe a set of locations within the warehouse 150 associated with each item included among its inventory. A location within a warehouse 150 associated with an item may correspond to a particular area within the warehouse 150 at which the item may be collected. For example, a layout of a warehouse 150 may describe one or more locations within the warehouse 150 associated with an item, in which each location corresponds to a particular aisle, a particular shelf, a particular display table, a particular promotional display, etc., at which the item may be collected.

[0059] In some embodiments, the online concierge system 102 further includes a route engine 324. The route engine

324 may access a layout of a warehouse **150** from the layout database **322** and identify a route for navigating from a location of a client device **110** within the warehouse **150** to one or more locations within the warehouse **150** associated with one or more items included among an inventory of the warehouse **150** to be collected during a shopping trip (e.g., items included in an order or a shopping list). The route engine **324** may do so based on the location of the client device **110** within the warehouse **150**, information identifying the item(s), and the layout of the warehouse **150**. For example, if a client device **110** is at a location within a warehouse **150** corresponding to an entrance/exit of the warehouse **150**, based on a layout of the warehouse **150**, the route engine **324** may identify a route for navigating from the entrance/exit to locations of items within the warehouse **150**, in which the items are included in a shopping list. In this example, the route may indicate an order in which the items are to be collected and locations along the route (e.g., specific shelves or locations within specific display cases) at which the items are located. In embodiments in which the layout database **322** maintains layouts of multiple warehouses **150**, a layout of a warehouse **150** accessed by the route engine **324** may be identified based on information identifying the warehouse **150** received by the route engine **324**, as further described below.

[0060] The route engine **324** may receive a location of a client device **110** within a warehouse **150** in various ways. In some embodiments, the route engine **324** may receive a location of a client device **110** within a warehouse **150** from a location tracking system **152** at the warehouse **150**. For example, a location tracking system **152** at a warehouse **150** may include one or more sensors **156** that detect client devices **110** scanning for Wi-Fi networks and record MAC addresses associated with any detected client devices **110**. In this example, the location tracking system **152** may track the location of a client device **110** (e.g., in various aisles or departments) as it moves throughout the warehouse **150** based on its MAC address and image or video data of a user of the client device **110** captured by one or more cameras included in a surveillance system **154** of the location tracking system **152**. Continuing with this example, the route engine **324** may receive the location of the client device **110** within the warehouse **150** from the location tracking system **152**. In some embodiments, the route engine **324** also or alternatively may receive other types of information from a location tracking system **152** at a warehouse **150** (e.g., information identifying the warehouse **150**, a time at which a client device **110** was at the warehouse **150** or at a specific location within the warehouse **150**, etc.). In the above example, the route engine **324** may receive the location of the client device **110** within the warehouse **150** in association with a warehouse identifier associated with the warehouse **150** and a timestamp indicating a date and time at which the client device **110** was at the location within the warehouse **150**. In various embodiments, the route engine **324** also or alternatively may receive a location of a client device **110** within a warehouse **150** from the item/location detection engine **330**. In such embodiments, the item/location detection engine **330** may detect the location of the client device **110** within the warehouse **150** based on a layout of the warehouse **150** and image or video data including a display area of the client device **110** received by the interface engine **328**, as further described below.

[0061] The route engine **324** also may receive information identifying a warehouse **150** and/or information identifying one or more items included among an inventory of the warehouse **150** to be collected during a shopping trip (e.g., one or more items included in an order or a shopping list). In some embodiments, the route engine **324** may receive information identifying a warehouse **150** from a location tracking system **152** at the warehouse **150**, as described above. The route engine **324** also or alternatively may receive information identifying a warehouse **150** from a client device **110** via GPS triangulation, cell tower triangulation, Wi-Fi positioning, or any other suitable technique. For example, if a client device **110** is at a warehouse **150**, the route engine **324** may track a location of the client device **110** (e.g., via GPS triangulation), access information describing the locations of various warehouses **150** (e.g., stored in the layout database **322** or a warehouse database), and identify the warehouse **150** based on the location of the client device **110**. The route engine **324** also or alternatively may receive information identifying a warehouse **150** and/or information identifying one or more items included among an inventory of the warehouse **150** to be collected during a shopping trip via an interface generated and transmitted by the interface engine **328**, as further described below. In embodiments in which a shopper **208** is fulfilling an order on behalf of a customer **204**, the route engine **324** may receive information identifying one or more items included in the order and/or information identifying a warehouse **150** from one or more components of the online concierge system **102**. For example, the route engine **324** may receive information identifying one or more items included in an order from the order fulfillment engine **306** and information identifying a warehouse **150** at which the order is to be fulfilled from the shopper management engine **310**.

[0062] The route engine **324** may identify a route using various techniques. In some embodiments, the route engine **324** may identify a route based on a distance from a location of a client device **110** within a warehouse **150** to one or more locations within the warehouse **150** associated with one or more items included among the inventory of the warehouse **150** to be collected during a shopping trip. For example, the route engine **324** may identify a route that requires a user of a client device **110** to travel a minimum distance to collect items included in a shopping list. In various embodiments, the route engine **324** also or alternatively may identify a route based on a predicted amount of time required to travel from a location of a client device **110** within a warehouse **150** to one or more locations within the warehouse **150** associated with one or more items included among the inventory of the warehouse **150** to be collected during a shopping trip. For example, based on information describing congestion/traffic within a warehouse **150** during different times of the day described in a layout of the warehouse **150**, the route engine **324** may identify a route that is predicted to take a user of a client device **110** a shortest amount of time to travel to collect items included in a shopping list.

[0063] In some embodiments, the route engine **324** also may identify a route using a machine-learned model. For example, the route engine **324** may identify a route using a machine-learned model that is trained based on information describing routes and other information associated with the routes (e.g., stored in the route database **326**, described below). In this example, the route engine **324** may provide an input to the machine-learned model including a location

of a client device **110** within a warehouse **150**, information identifying one or more items to be collected during a shopping trip, and a layout of the warehouse **150**. Continuing with this example, the route engine **324** may receive an output from the model identifying a route that requires a user of the client device **110** to travel a minimum distance to collect the item(s) and/or a route that is predicted to take the user a shortest amount of time to collect the item(s).

[0064] In various embodiments, the route engine **324** also may modify a route based on a location within a warehouse **150** associated with a replacement item. For example, suppose that a user of a client device **110** has followed a route identified by the route engine **324** and is at a location within a warehouse **150** associated with an item included in a shopping list and the item is out of stock. In this example, once a replacement item is identified (e.g., by a customer **204** or by the item replacement engine **332** and/or the experience enhancement engine **338**, described below), the route engine **324** may identify a location within the warehouse **150** associated with the replacement item based on a layout of the warehouse **150**. In the above example, the route engine **324** may modify the route such that the modified route is for navigating from the location associated with the item to the location associated with the replacement item and the locations associated with any remaining items in the shopping list.

[0065] In some embodiments, the route engine **324** also may modify a route identified by the route engine **324** if a user of a client device **110** deviates from the route. In such embodiments, the route engine **324** may modify the route based on a current location of the client device **110** within the warehouse **150**. For example, if a user of a client device **110** deviates from a route identified by the route engine **324** after collecting some items included in a shopping list, the route engine **324** may receive information from the item/location detection engine **330** (described below) indicating the user has deviated from the route and a current location of the client device **110** within a warehouse **150**. In this example, the route engine **324** may modify the route based on the current location of the client device **110** within the warehouse **150** and the locations associated with the remaining items in the shopping list, such that the items may be collected in an order that is different from that associated with the original route. In embodiments in which the route engine **324** modifies a route, the route engine **324** may do so using the same techniques described above for identifying a route.

[0066] In various embodiments, the route engine **324** may be configured to store information describing routes (i.e., routes identified or modified by the route engine **324**) in a route database **326**. Information describing a route may be stored in the route database **326** in association with various types of information. Information describing a route may be stored in association with information describing a client device **110** associated with the route (e.g., an IP address associated with the client device **110**), a user associated with the client device **110** (e.g., a username, email address, or other identifier associated with the user), and/or a warehouse **150** associated with the route (e.g., a warehouse identifier). Information describing a route also or alternatively may be stored in association with a time associated with the route (e.g., a timestamp indicating when it was identified or modified by the route engine **324**), information identifying items to be collected along the route, a distance required to

travel the route, an amount of time (predicted or actual) required to travel the route, or any other suitable types of information that may be associated with a route.

[0067] In some embodiments, the online concierge system **102** further includes an interface engine **328**. The interface engine **328** generates augmented reality elements that are overlaid onto portions of display areas of client devices **110** to which the augmented reality elements are sent. For example, an augmented reality element may be overlaid onto a portion of a display area of a client device **110** at which an item is detected (e.g., by the item/location detection engine **330**, described below). Alternatively, in the above example, the augmented reality element may be overlaid onto another portion of the display area, such that it appears to float in a corner of the display area or near the item detected within the display area.

[0068] In some embodiments, the interface engine **328** generates augmented reality elements that include instructions for navigating routes (i.e., routes identified or modified by the route engine **324**) within one or more warehouses **150**. In such embodiments, the augmented reality elements may include turn-by-turn instructions for navigating the routes. For example, based on a route identified by the route engine **324**, the interface engine **328** may generate augmented reality elements that correspond to arrows that, when overlaid onto a portion of a display area of a client device **110** at which the floor of a warehouse **150** is detected (e.g., by the item/location detection engine **330**, described below), provide instructions for navigating the route. In this example, a user of the client device **110** may follow the arrows while at the warehouse **150** to navigate to multiple locations associated with items included in a shopping list. In the above example, if the user of the client device **110** deviates from the route, the route engine **324** may modify the route based on the current location of the client device **110** and the interface engine **328** may generate another set of augmented reality elements that correspond to arrows that provide instructions for navigating the modified route.

[0069] In some embodiments, the interface engine **328** generates augmented reality elements that call attention to items included among an inventory of a warehouse **150** and/or locations within the warehouse **150** associated with the items, in which the items and/or locations are detected within a display area of a client device **110**. In such embodiments, an augmented reality element may call attention to an item or a location by highlighting it (e.g., with one or more colors) or with icons (e.g., arrows or circles), animations, etc. For example, an augmented reality element may call attention to an item detected within a display area of a client device **110** by encircling a portion of the display area at which the item is detected with a colored halo. As an additional example, if a location within a warehouse **150** (e.g., an area on a shelf) associated with an item is detected within a display area of a client device **110**, an augmented reality element may call attention to the location with a flashing arrow that points to a portion of the display area at which the location is detected. An augmented reality element may call attention to an item for various reasons (e.g., if the item is included in an order or a shopping list, if the item is a replacement item, if one or more attributes of the item match profile information associated with a customer **204**, etc.). For example, if a set of preferences associated with a customer **204** includes a dietary restriction indicating that the customer **204** is allergic to pineapple, an augmented

reality element may call attention to an item that contains pineapple with a warning icon that hovers over or near a portion of a display area of a client device **110** at which the item is detected. In this example, if the set of preferences associated with the customer **204** also indicates that the customer **204** likes strawberries, another augmented reality element may call attention to strawberry-flavored candy with a heart icon that hovers over or near a portion of the display area of the client device **110** at which the item is detected. As yet another example, an augmented reality element may call attention to an item that is included in an order or a shopping list by encircling a portion of a display area of a client device **110** at which the item is detected with a colored halo. Similarly, an augmented reality element may call attention to a location within a warehouse **150** associated with an item if the item is included in an order or a shopping list, if the item is a replacement item, if one or more attributes of the item match one or more portions of profile information associated with a customer **204**, etc.

[0070] In various embodiments, the interface engine **328** also generates augmented reality elements that call attention to attributes of items (e.g., sizes, varieties, brands, etc.) included among an inventory of a warehouse **150** detected within a display area of a client device **110**. In some embodiments, an augmented reality element may call attention to an attribute of an item by highlighting it (e.g., with one or more colors) or with icons (e.g., arrows or circles), animations, etc. An augmented reality element may call attention to an attribute of an item for various reasons (e.g., if an item having the attribute is included in an order or a shopping list, if the attribute matches profile information associated with a customer **204**, etc.). For example, suppose that profile information stored in the customer database **314** indicates that a customer **204** is allergic to soy and the item/location detection engine **330** (described below) detects that the customer **204** is holding an item that contains soy. In this example, the experience enhancement engine **338** (also described below) may infer that the customer **204** is thinking about purchasing the item and may match the soy ingredient with the profile information indicating the customer **204** is allergic to soy. Continuing with this example, based on the inference, the interface engine **328** may generate an augmented reality element that calls attention to the word “soy” in a list of ingredients on the item detected within a display area of a client device **110** with a warning icon and by encircling a portion of the display area in which the word is detected with a red halo. As an additional example, if an item (e.g., ground coffee) is included in a shopping list, an augmented reality element may call attention to information describing an attribute of the item (e.g., the word “ground” on the item) detected within a display area of a client device **110** by encircling a portion of the display area in which the information is detected with a green halo. In various embodiments, multiple augmented reality elements may call attention to different attributes by highlighting them differently (e.g., with different colors) or with different icons, different animations, etc. In the above example, if another item (e.g., whole bean coffee) is not included in the shopping list, another augmented reality element may call attention to information describing an attribute of the item (e.g., the word “whole bean” on the item) detected within the display area of the client device **110** by encircling a portion of the display area in which the information is detected with a red halo.

[0071] In some embodiments, the interface engine **328** may update an augmented reality element that calls attention to something. For example, if the item/location detection engine **330** (described below) determines that an item is not at a location within a warehouse **150** associated with the item, the interface engine **328** may update an augmented reality element that calls attention to the location (e.g., by graying it out, muting it, etc.). In some embodiments, an augmented reality element that calls attention to something may be updated if a user of a client device **110** adjusts one or more settings to change an appearance of the augmented reality element (e.g., by changing one or more colors of highlights, by changing the style of icons or animations, etc.).

[0072] In some embodiments, the interface engine **328** generates augmented reality elements that obscure items included among an inventory of a warehouse **150** detected within a display area of a client device **110**. In such embodiments, an augmented reality element may obscure an item by blocking it partially or completely from view, blurring it, etc. For example, an augmented reality element, such as a colored rectangle or other shape that is mostly or completely opaque, may obscure an item if it is overlaid onto a portion of a display area of a client device **110** at which the item is detected. An augmented reality element may obscure an item for various reasons (e.g., if profile information associated with a customer **204** indicates that the customer **204** dislikes the item/attributes of the item, if the profile information indicates the customer **204** has a restriction, such as a dietary restriction, associated with the item, etc.). For example, suppose that the item/location detection engine **330** (described below) detects an advertisement for an item within a display area of a client device **110** and the experience enhancement engine **338** (also described below) matches an attribute of the item with information stored in the customer database **314** indicating that a customer **204** dislikes items with the attribute. In this example, the interface engine **328** may generate an augmented reality element that includes a coupon for a different item that does not have the attribute, in which the augmented reality element may obscure the advertisement if it is overlaid onto a portion of the display area at which the advertisement is detected. In some embodiments, a user of a client device **110** may adjust one or more settings to change an augmented reality element that obscures an item (e.g., by changing the method by which it obscures the item, by changing its opacity, etc.).

[0073] In various embodiments, the interface engine **328** generates augmented reality elements that compare items included among an inventory of a warehouse **150** detected within a display area of a client device **110**. In such embodiments, an augmented reality element that compares items may include statistics associated with each item or other values of one or more attributes of each item. An augmented reality element may compare two or more items for various reasons (e.g., if a user of a client device **110** is holding the items, if the items are detected within a display area of a client device **110** for at least a threshold amount of time, etc.). For example, an augmented reality element that compares two sports drinks being held by a user of a client device **110** may include side-by-side statistics associated with the drinks (e.g., sugar content, number of calories, and price per ounce). Furthermore, an augmented reality element that compares items may highlight certain statistics, such as

statistics related to profile information associated with a customer **204** that the experience enhancement engine **338** (described below) matches with one or more attributes of the items. In the above example, if the customer **204** is on a diet that limits their sugar intake, the augmented reality element may highlight the sugar content of the drinks. In some embodiments, a user of a client device **110** may adjust one or more settings to change an augmented reality element that compares items, such as how statistics or other values are displayed (e.g., numerically in a table or graphically in a chart), by changing the types of values displayed, etc.

[0074] The interface engine **328** also may generate augmented reality elements that recommend items included among an inventory of a warehouse **150** detected within a display area of a client device **110**. An augmented reality element that recommends an item may include information associated with the item (e.g., brand name, item type, size, nutritional information, price, discounts or coupons that may apply to the item, etc.) and a reason the item is being recommended. An item may be recommended if a customer **204** is likely to purchase it (e.g., based on their order/purchase history or other profile information, other items included in an order or shopping list, etc.), if it is a replacement item for another item in a customer's order or shopping list that is out of stock, if it is a replacement item the customer **204** is likely to prefer to another item (e.g., in an order or a shopping list), etc. For example, if an item in a shopping list is out of stock, an augmented reality element may indicate this and recommend a replacement item that is similar to the one that is out of stock. In this example, the augmented reality element may include a brand of the replacement item, a description of the replacement item, and a price of the replacement item. In some embodiments, a user of a client device **110** may adjust one or more settings to change augmented reality elements that recommend items, such as types of information about the items being recommended (e.g., unit price, item price, brand, etc.), criteria for recommending items, etc.

[0075] In various embodiments, the interface engine **328** generates augmented reality elements that promote items included among an inventory of a warehouse **150** detected within a display area of a client device **110**. In such embodiments, an augmented reality element may be generated based on content associated with the item (e.g., stored in the content database **334**). An augmented reality element may promote an item for various reasons (e.g., it is a new item, a customer **204** is considering purchasing it, it is on sale or associated with a promotion, etc.). In some embodiments, the interface engine **328** generates augmented reality elements that promote items based on promotional content (e.g., advertisements, coupons, deals, etc.) associated with the items. For example, the interface engine **328** may generate an augmented reality element promoting an item that includes an advertisement or a promotion associated with the item (e.g., buy two, get one free). In various embodiments, the interface engine **328** generates augmented reality elements that promote items based on content describing how the items may be used. For example, if an item is a food item, the interface engine **328** may generate an augmented reality element promoting the item with a recipe that includes the item as an ingredient. As an additional example, if an item is an article of clothing or an accessory, the interface engine **328** may generate an aug-

mented reality element promoting the item that includes an image or a video clip of a person modeling the item.

[0076] In some embodiments, the interface engine **328** generates an augmented reality element that includes a list of items included among an inventory of a warehouse **150**. A list may include one or more items identified by a customer **204** (e.g., in an order received from the customer **204**) or by a user of a client device **110** (e.g., in a shopping list received in an input via an interface generated and transmitted by the interface engine **328**, as described below). For example, if a customer **204** places a delivery order via the online concierge system **102**, the interface engine **328** may generate an augmented reality element that includes a list of items included in the order. In this example, the augmented reality element subsequently may be sent to a display area of a client device **110** associated with a shopper **208** assigned to fulfill the order. As an additional example, if information identifying items included in a shopping list is received in an input from a user of a client device **110** via the interface generated and transmitted by the interface engine **328**, the interface engine **328** may generate an augmented reality element that includes the shopping list. In some embodiments, a list of items included in an augmented reality element may be ordered and/or numbered based on a route identified by the route engine **324**. For example, a list of items included in an augmented reality element may be ordered and numbered based on an order in which they will be encountered along a route identified by the route engine **324**.

[0077] In some embodiments, if the interface engine **328** generates an augmented reality element that includes a list of one or more items included among an inventory of a warehouse **150**, the augmented reality element may be updated. In such embodiments, the interface engine **328** may update the augmented reality element if an item is unavailable, if an item is collected, if a replacement item is identified, etc. For example, if the interface engine **328** generates an augmented reality element that includes a shopping list, as each item in the shopping list is collected, the interface engine **328** may update the augmented reality element (e.g., by checking a checkbox next to each collected item, by crossing each collected item off the list, by removing each collected item from the list, etc.). In the above example, if the item/location detection engine **330** (described below) determines that one of the items is out of stock, the interface engine **328** may update the augmented reality element so that it indicates that the item is not in stock (e.g., by graying it out and adding a note indicating it is out of stock). Continuing with the above example, if the item replacement engine **332** (described below) subsequently identifies a replacement item, the interface engine **328** may update the augmented reality element so that it includes the replacement item and a note indicating it is a replacement item.

[0078] In various embodiments, the interface engine **328** also may generate an augmented reality element that includes a route for navigating from a location of a client device **110** within a warehouse **150** to one or more locations within the warehouse **150** associated with one or more items identified by a customer **204** to be collected during a shopping trip (e.g., items included in an order or a shopping list). An augmented reality element that includes a route may indicate a location of a client device **110** within a warehouse **150**, locations within the warehouse **150** along the route at

which one or more items identified by a customer **204** may be collected, and an order in which the item(s) may be collected. For example, an augmented reality element that includes a route may depict a layout of a warehouse **150**, in which a location of a client device **110** and locations associated with items included in an order or shopping list are represented by icons that are connected by a line representing the route. In this example, the locations associated with the items may be numbered based on an order in which a user of the client device **110** will encounter the locations associated with the items by following the route. In some embodiments, an augmented reality element that includes a route may be updated (e.g., as a client device **110** moves within a warehouse **150**, as items identified by a customer **204** are collected, as the route is modified by the route engine **324**, etc.). In the above example, as the user of the client device **110** moves along the route, the icon representing the location of the client device **110** also moves, tracking the location of the client device **110** within the warehouse **150**. In this example, as items are collected, icons representing the locations associated with the items may be removed, grayed out, checked off, etc.

[0079] Once the interface engine **328** generates or updates augmented reality elements, the interface engine **328** may send them to display areas of client devices **110**. As described above, augmented reality elements are overlaid onto portions of display areas of client devices **110** to which the augmented reality elements are sent. An augmented reality element may be overlaid onto a portion of a display area of a client device **110** based on a location within the display area at which an item is detected (e.g., by the item/location detection engine **330**, described below). For example, an augmented reality element may be overlaid onto a portion of a display area of a client device **110** at which an item is detected. Alternatively, in the above example, the augmented reality element may be overlaid onto another portion of the display area, such that it appears to float in a corner of the display area or near the item detected within the display area. In various embodiments, as physical elements, locations, etc. detected within a display area of a client device **110** move within the display area, one or more augmented reality elements may move as well. For example, if an augmented reality element calls attention to an item, as the item moves within a display area of a client device **110**, the augmented reality element may move as well, such that it tracks the movement of the item. In some embodiments, augmented reality elements may be hidden by a user of a client device **110**, such that they are not visible within a display area of the client device **110**. For example, a user of a client device **110** may switch between different modes, such that in one mode, augmented reality elements are not visible within a display area of the client device **110**, while in another mode, some or all of the augmented reality elements are visible within the display area of the client device **110**.

[0080] In some embodiments, the interface engine **328** may store information describing augmented reality elements sent to client devices **110** in the customer database **314**, such that this information is included among profile information associated with customers **204** associated with the client devices **110**. In such embodiments, the customers **204** may be users of the client devices **110** or may have placed orders fulfilled by shoppers **208** who were using the client devices **110** when the augmented reality elements

were sent. For example, suppose that the interface engine **328** sends an augmented reality element to a display area of a client device **110** associated with a customer **204**, in which the augmented reality element promotes an item and includes a content item (e.g., a coupon) associated with the item. In this example, information describing the augmented reality element (information indicating that it promoted the item, information identifying the item and the content item, etc.) and a date and time that it was sent to the display area of the client device **110** may be stored among profile information associated with the customer **204** in the customer database **314**.

[0081] The interface engine **328** also may generate and transmit an interface that includes one or more fields, buttons, or other interactive elements for receiving various inputs from a user of a client device **110**. In some embodiments, the interface engine **328** may receive information identifying a warehouse **150** from a user of a client device **110** via the interface. For example, if a user of a client device **110** is at a warehouse **150**, the user may select an option to share the location of the client device **110** by clicking on a button included in the interface. Alternatively, in the above example, the user may select the warehouse **150** from a drop-down menu included in the interface or type a name of a retailer associated with the warehouse **150** into a search box included in the interface and select the warehouse **150** from a list of results. In some embodiments, via the interface, the interface engine **328** also or alternatively may receive information from a user of a client device **110** identifying one or more items included among an inventory of a warehouse **150** to be collected from the warehouse **150** during a shopping trip. In such embodiments, upon receiving the input from the user via the interface, the interface engine **328** may retrieve information identifying items included among the inventory of the warehouse **150** based on the input (e.g., items having attributes with values that match at least a portion of the input identified by leveraging item embeddings). Via the interface, one or more items may then be added to a list of items (e.g., a shopping list) or subsequently removed from the list. For example, the interface may include a search box that allows a user to type in a name of an item, a description of an item, etc. In this example, the user may add one or more items included in a list of results to a shopping list and search for additional items to add to the shopping list.

[0082] In some embodiments, the interface generated and transmitted by the interface engine **328** also may receive inputs associated with one or more augmented reality elements generated by the interface engine **328**. In such embodiments, the interface engine **328** may receive various requests from a user of a client device **110** via the interface (e.g., to accept a replacement item recommended in an augmented reality element, to adjust one or more settings associated with an augmented reality element, to check off items in a shopping list included in an augmented reality element, to claim an offer presented in an augmented reality element, etc.). For example, the interface engine **328** may receive a request from a user of a client device **110** to switch between different modes via the interface, in which different modes change the visibility of augmented reality elements to the user. As an additional example, if a replacement item is recommended to a user of a client device **110**, the interface engine **328** may receive a request from the user to accept the replacement item via the interface and the interface engine

328 may communicate the acceptance to the route engine **324** so that the route may be modified based on a location associated with the replacement item. As yet another example, if a shopping list is included in an augmented reality element, as items included in the shopping list are collected, the interface engine **328** may receive requests from a user of a client device **110** to check the collected items off the list via the interface.

[0083] The interface engine **328** also may receive image or video data captured by a camera of a client device **110**, in which the image or video data includes a display area of the client device **110**. For example, if a client device **110** is a smartphone, a camera of the smartphone may capture image or video data that is visible within a display area of the smartphone to a user of the smartphone. As an additional example, if a client device **110** is a pair of augmented reality glasses, a camera of the augmented reality glasses may capture image or video data that is visible to a user wearing the augmented reality glasses. The interface engine **328** may communicate image or video data it receives to the item/location detection engine **330**, which may then detect various physical elements (e.g., items), positions of physical elements, locations, etc. within a display area of a client device **110**, as described below.

[0084] In some embodiments, the online concierge system **102** further includes an item/location detection engine **330**. The item/location detection engine **330** may detect various physical elements (e.g., items included among an inventory of a warehouse **150**, building elements, organizational elements, shopping carts or baskets, advertisements, etc.) within a display area of a client device **110**. In some embodiments, the item/location detection engine **330** may do so based on images of the physical elements and image or video data captured by a camera of a client device **110**. In such embodiments, images of the physical elements may include images of items included in a catalog stored in the inventory database **304**, images of organizational elements and building elements included in a layout stored in the layout database **322**, images of advertisements or other types of promotional content stored in the content database **334**, etc. For example, suppose that video data captured by a camera of a client device **110** received by the interface engine **328** depicts a physical element, such as a box of cereal, displayed within a display area of the client device **110**. In this example, the interface engine **328** may communicate the video data to the item/location detection engine **330**, which may access images of items included in a catalog stored in the inventory database **304** and compare the depiction to the images of the items included in the catalog. In the above example, the item/location detection engine **330** may detect the box of cereal if the depiction has at least a threshold measure of similarity to one or more images of the box of cereal included in the catalog.

[0085] In some embodiments, the item/location detection engine **330** may detect physical elements using a machine-learned model. For example, the item/location detection engine **330** may provide image or video data as an input to a machine-learned model that is trained using images of various physical elements within one or more warehouses **150** (e.g., stored in the inventory database **304**, the layout database **322**, the content database **334**, etc.). The item/location detection engine **330** may then receive an output from the model describing predicted likelihoods that each physical element is detected in the image or video data and

determine that a physical element is detected if the corresponding predicted likelihood is at least a threshold likelihood. Alternatively, the item/location detection engine **330** may determine that the physical element is not detected if the corresponding predicted likelihood is less than the threshold likelihood.

[0086] In various embodiments, the item/location detection engine **330** may detect positions of physical elements relative to each other. In some embodiments, the item/location detection engine **330** may detect that a physical element is within another physical element, that a physical element is resting or stacked onto another physical element, etc. For example, if the item/location detection engine **330** detects an item included among an inventory of a warehouse **150** and a shopping basket, the item/location detection engine **330** also may detect that the item is within the shopping basket. In various embodiments, the item/location detection engine **330** may detect that items are being held by users of client devices **110** (e.g., based on the positions of the items relative to the users' hands). The item/location detection engine **330** may do so using various techniques, such as machine learning/object detection, depth perception, infrared/thermal imaging, or any other suitable technique or combination of techniques. For example, suppose that a client device **110** includes multiple infrared cameras, such that the client device **110** is capable of receiving depth perception information and heat-related information about an item detected within a display area of the client device **110** and other physical elements detected within the display area. In this example, the item/location detection engine **330** may detect that the item is being held if the item and a hand holding the item are moving together in directions and/or at speeds that are different from all or most other physical elements detected within the display area. In the above example, the item/location detection engine **330** also may detect that the item is being held if depth perception information indicates that the item and hand are closer to the client device **110** than the other physical elements and if heat-related information indicates that the item and hand emit different amounts of heat.

[0087] The item/location detection engine **330** also may detect a location and/or an orientation of a client device **110** within a warehouse **150**. In such embodiments, the item/location detection engine **330** may receive (e.g., via the interface engine **328**) image or video data including a display area of the client device **110** captured by a camera of the client device **110** and detect the location and/or orientation of the client device **110** based on the image or video data and the layout of the warehouse **150**. For example, if a client device **110** is at a warehouse **150**, the item/location detection engine **330** may receive image or video data captured by a camera of the client device **110** from the interface engine **328**, in which the image or video data includes various physical elements within the warehouse **150**. In this example, the item/location detection engine **330** may then compare the physical elements included in the image or video data to those of various locations within the warehouse **150** included in a layout of the warehouse **150**. Continuing with this example, the item/location detection engine **330** may then detect a location and/or orientation within the warehouse **150** corresponding to the location and/or orientation of the client device **110** if the physical

elements included in the image or video data have at least a threshold measure of similarity to the physical elements at the location.

[0088] In various embodiments, the item/location detection engine 330 may detect that a user of a client device 110 has deviated from a route. In such embodiments, the item/location detection engine 330 may do so based on the route, a layout of a warehouse 150, and image or video data captured by a camera of the client device 110. For example, the item/location detection engine 330 may detect a location of a client device 110 within a warehouse 150 based on a layout of the warehouse 150 and physical elements detected within a display area of the client device 110, as described above. In this example, the item/location detection engine 330 may detect that a user of the client device 110 has deviated from a route if the detected location does not correspond to a location within the warehouse 150 along the route. In some embodiments, upon detecting that a user of a client device 110 has deviated from a route, the item/location detection engine 330 may communicate this to the route engine 324 in association with information describing a current location of the client device 110 and the route engine 324 may then modify the route.

[0089] In some embodiments, a location within a warehouse 150 detected within a display area of a client device 110 by the item/location detection engine 330 may be associated with an item included among an inventory of the warehouse 150. For example, suppose that an item is associated with a particular shelf within a warehouse 150 at which the item may be collected. In this example, the item/location detection engine 330 may receive image or video data captured by a camera of a client device 110 from the interface engine 328 and compare physical elements included in the image or video data to those at the location associated with the item included in a layout of the warehouse 150. Continuing with this example, the item/location detection engine 330 may detect the location within the warehouse 150 associated with the item if the physical elements included in the image or video data have at least a threshold measure of similarity to the physical elements (i.e., the particular shelf and any surrounding physical elements) at the location associated with the item.

[0090] In various embodiments, once the item/location detection engine 330 has detected a location within a warehouse 150 associated with an item, the item/location detection engine 330 may determine whether the item is at the location. In some embodiments, the item/location detection engine 330 may do so based on one or more images of the item (e.g., one or more images included in a catalog stored in the inventory database 304) and image or video data captured by a camera of a client device 110 received from the interface engine 328. For example, the item/location detection engine 330 may receive image or video data captured by a camera of a client device 110 from the interface engine 328, in which the image or video data includes various physical elements displayed within the display area of the client device 110. In this example, once the item/location detection engine 330 has detected a location within a warehouse 150 associated with an item, the item/location detection engine 330 may access images of the item included in a catalog stored in the inventory database 304 and compare the physical elements included in the image or video data to the images of the item included in the catalog. Continuing with this example, if the physical ele-

ments have at least a threshold measure of similarity to the images of the item included in the catalog, the item/location detection engine 330 may determine the item is at the location associated with the item. However, in the above example, if the physical elements do not have at least the threshold measure of similarity to the images of the item included in the catalog, the item/location detection engine 330 may determine the item is not at the location associated with the item. In some embodiments, the item/location detection engine 330 may determine whether an item is at a location within a warehouse 150 associated with the item using a machine-learned model, as described above.

[0091] In some embodiments, once the item/location detection engine 330 has detected a location within a warehouse 150 associated with an item and determines that the item is not at the location, the item/location detection engine 330 also may determine whether the item is detected elsewhere within the warehouse 150. The item/location detection engine 330 may make this determination within a threshold amount of time of detecting the location associated with the item, within a threshold distance of detecting the location associated with the item, etc. For example, suppose that an item is misplaced on a shelf above or below a specific shelf corresponding to a location within a warehouse 150 associated with the item. In this example, once the item/location detection engine 330 has detected the location within the warehouse 150 associated with the item and determines that the item is not at the location, the item/location detection engine 330 may determine whether the item is detected within a display area of a client device 110 while the location is also detected within the display area.

[0092] In some embodiments, the item/location detection engine 330 may communicate information describing changes in inventory for a warehouse 150 to the inventory database 304, which may update inventory information associated with the warehouse 150 based on the information. In some embodiments, if the item/location detection engine 330 determines that an item is not at a location within a warehouse 150 associated with the item (or elsewhere within the warehouse 150, in some embodiments), the item/location detection engine 330 may communicate this to the inventory management engine 302. The inventory management engine 302 may then update inventory information associated with the warehouse 150 maintained in the inventory database 304 to reflect that the item is out of stock at the warehouse 150. In some embodiments, the item/location detection engine 330 also may detect a quantity of an item at a warehouse 150. In such embodiments, the item/location detection engine 330 may communicate the quantity to the inventory management engine 302, which may then update the inventory database 304 to reflect the quantity of the item detected at the warehouse 150.

[0093] Furthermore, in embodiments in which the item/location detection engine 330 detects a quantity of an item at a warehouse 150, the item/location detection engine 330 may update a layout of the warehouse 150 (e.g., stored in the layout database 322). In some embodiments, the item/location detection engine 330 may update a layout of a warehouse 150 if at least a threshold quantity of an item is detected at the warehouse 150 and the item/location detection engine 330 determines that the threshold quantity of the item is not at a location within the warehouse 150 associated with the item. For example, suppose that the item/location detection engine 330 detects at least ten cans of a particular

brand and type of soup and determines that the cans of soup are not at a location within a warehouse **150** associated with the can of soup based on a layout of the warehouse **150**. In this example, the item/location detection engine **330** may update the layout of the warehouse **150** to reflect that the location within the warehouse **150** at which the cans of soup were detected is associated with the particular brand and type of soup.

[0094] In some embodiments, the online concierge system **102** further includes an item replacement engine **332**. The item replacement engine **332** identifies replacement items included among an inventory of a warehouse **150** that may replace other items included among the inventory of the warehouse **150** (e.g., if the items that may be replaced are not available, if a customer **204** is likely to prefer the replacement items to the items that may be replaced, etc.). The item replacement engine **332** may do so by accessing a set of attributes of an item that may be replaced and attributes of other items included among the inventory of a warehouse **150** (e.g., stored in the inventory database **304**) and identifying one or more replacement items that each have a set of attributes that have at least a threshold measure of similarity to the set of attributes of the item that may be replaced. In various embodiments, the item replacement engine **332** may identify a replacement item by accessing a taxonomy maintained by the inventory management engine **302** described above and identifying a set of potential replacement items included in a most specific category in which the item that may be replaced is included. The item replacement engine **332** may then compare a set of attributes of each potential replacement item with a set of attributes of the item that may be replaced and identify a replacement item from the set of potential replacement items based on the comparison, in which the set of attributes of the replacement item has at least a threshold measure of similarity to the set of attributes of the item that may be replaced.

[0095] In some embodiments, replacement items also may be identified in other ways. In embodiments in which a shopper **208** is fulfilling a delivery order on behalf of a customer **204**, replacement items may be identified by the customer **204**. For example, if a shopper **208** is fulfilling an order on behalf of a customer **204** and an item, such as a particular brand and flavor of ice cream included in the order is not available, the shopper **208** may identify potential replacement items that have at least a threshold measure of similarity to the item (e.g., other flavors of ice cream of the same brand or other brands of ice cream of the same flavor). In this example, the shopper **208** may communicate the potential replacement items to a client device **110** associated with the customer **204** (e.g., by capturing an image or a video of them and sending the image or video via text or email, by listing or describing them during a phone call or in a text message, etc.). Continuing with this example, the shopper **208** may then receive a response from the customer **204** identifying a replacement item included among the potential replacement items. In various embodiments, the item replacement engine **332** may identify a set of potential replacement items that each have a set of attributes that have at least a threshold measure of similarity to a set of attributes of an item that may be replaced. The item replacement engine **332** may do so by accessing attributes of the item that may be replaced and attributes of other items included among the inventory of a warehouse **150** or by accessing a taxonomy maintained by the inventory management engine

302, as described above. In such embodiments, the item replacement engine **332** may then communicate information identifying the set of potential replacement items (and information identifying the item that may be replaced, in some embodiments) to the experience enhancement engine **338**, which may then identify one or more replacement items from the set of potential replacement items, as described below.

Personalization Components

[0096] In some embodiments, the online concierge system **102** further includes a content database **334**. The content database **334** may store various types of content associated with items included among an inventory of a warehouse **150**. Examples of types of content that may be associated with an item include promotional content, content describing how the item may be used, or any other types of content that may be associated with an item. For example, promotional content associated with an item may include advertisements for the item, coupons or deals associated with the item, offers associated with the item (e.g., free samples), etc. As an additional example, content describing how an item may be used may include a recipe that includes the item as an ingredient, an image or a video demonstrating how the item may be used, etc. Content stored in the content database **334** may be obtained from users, third-party systems **130** (e.g., websites or applications), or any other suitable source, and stored in the content database **334**. Additionally, each content item stored in the content database **334** may have one or more attributes describing the content. Examples of attributes of a content item include a type associated with the content item (e.g., coupon, advertisement, recipe, etc.), information identifying one or more items associated with the content item, attributes of each item associated with the content item, one or more dates associated with the content item (e.g., an expiration date associated with a coupon or offer), a type of data included in the content item (e.g., image data, video data, text data, etc.), or any other suitable types of information. Attributes of a content item may be provided by a source from which the content item was received or may be determined by the online concierge system **102** from information included in the content item.

[0097] Additionally, in some embodiments, the content database **334** maintains a content graph identifying connections between content items stored in the content database **334**. A connection between content items indicates that the connected content items each have one or more common attributes. In some embodiments, a connection between content items indicates that a customer **204** included items associated with each connected content item in a common order or included items associated with each connected content item in orders the online concierge system **102** received from the customer **204** within a threshold amount of time of each other. In various embodiments, each connection between content items includes a value, with the value providing an indication of a strength of a connection between the content items.

[0098] In some embodiments, the online concierge system **102** further includes a profile update engine **336**. The profile update engine **336** may update profile information associated with a customer **204** stored in the customer database **314**. In some embodiments, the profile update engine **336** may update a set of preferences associated with a customer **204** stored in the customer database **314** based on other

profile information associated with the customer 204. For example, suppose that profile information associated with a customer 204 indicates that the customer 204 has a goal of losing weight by following a particular diet. In this example, the profile update engine 336 may update a set of preferences associated with the customer 204 indicating that the customer 204 has a preference for food items that may be consumed when on the diet (e.g., foods low in sugar and saturated fat) and a dislike for food items that are to be avoided when on the diet (e.g., foods high in sugar and saturated fat).

[0099] In some embodiments, the online concierge system 102 further includes an experience enhancement engine 338. The experience enhancement engine 338 may communicate various types of information to the interface engine 328, which may then generate augmented reality elements based on this information. In some embodiments, this information may include inferences made by the experience enhancement engine 338, information describing replacement items identified by the experience enhancement engine 338, and/or information describing profile or other information associated with customers 204 that the experience enhancement engine 338 matches with attributes of items or other physical elements. In various embodiments, this information also may include information the experience enhancement engine 338 accesses from various databases (e.g., the inventory database 304, the customer database 314, the content database 334, etc.).

[0100] In some embodiments, the experience enhancement engine 338 infers an intent of a user of a client device 110 based on information detected by the item/location detection engine 330 and communicates the inference to the interface engine 328, which then generates an augmented reality element based on the inference. In various embodiments, the experience enhancement engine 338 also may communicate an inference to the interface engine 328 in association with information retrieved from one or more databases, which the interface engine 328 also may use to generate an augmented reality element. For example, if the item/location detection engine 330 detects that a user of a client device 110 is holding an item or if the item has been detected within a display area of the client device 110 for at least a threshold amount of time, the experience enhancement engine 338 may infer that the user is thinking about purchasing the item. In this example, the experience enhancement engine 338 may retrieve promotional content (e.g., a coupon and an advertisement) associated with the item from the content database 334 and communicate the inference and the promotional content to the interface engine 328, which may then generate an augmented reality element that promotes the item by including the promotional content. Alternatively, in the above example, the experience enhancement engine 338 may retrieve a set of content from the content database 334 describing how the item may be used (e.g., a recipe that includes the item as an ingredient) and communicate the content to the interface engine 328, which may then generate an augmented reality element that promotes the item by including the content. As an additional example, if the item/location detection engine 330 detects that a user of a client device 110 is holding two or more items or if the items have been detected within a display area of the client device 110 for at least a threshold amount of time, the experience enhancement engine 338 may infer that the user is comparing the items. In this example, the experience enhancement engine 338 may access attributes of the

items stored in the inventory database 304 and communicate the inference and the attributes to the interface engine 328, which may then generate an augmented reality element that compares the items based on the inference and attributes.

[0101] In some embodiments, the experience enhancement engine 338 also may infer an intent of a customer 204 based on items included in an order or shopping list associated with the customer 204 and content stored in the content database 334 (e.g., recipes). In such embodiments, the experience enhancement engine 338 may then communicate the inference and any information related to the inference (e.g., retrieved from the inventory database 304, the content database 334, etc.) to the interface engine 328, which may generate an augmented reality element based on the inference and information. For example, suppose that two items (e.g., graham crackers and marshmallows) that are ingredients in a popular recipe stored in the content database 334 that calls for three ingredients are included in an order placed by a customer 204. In this example, if the last ingredient (e.g., chocolate) is not included in the order, the experience enhancement engine 338 may infer that the customer 204 is likely to purchase this item as well and may send information describing the inference and the item (e.g., retrieved from the inventory database 304) to the interface engine 328. In the above example, the interface engine 328 may then generate an augmented reality element recommending the item, which is subsequently sent to a display area of a client device 110 associated with a shopper 208 assigned to fulfill the customer's order. Continuing with this example, the shopper 208 may then communicate the recommendation to the customer 204.

[0102] In embodiments in which the experience enhancement engine 338 communicates information stored in various databases (e.g., the inventory database 304, the content database 334, etc.) to the interface engine 328, which then generates augmented reality elements based on the information, the experience enhancement engine 338 may identify the information to be communicated based on profile information stored in the customer database 314. In some embodiments, the experience enhancement engine 338 may identify the information to be communicated based on profile information associated with customers 204 describing augmented reality elements sent to display areas of client devices 110 associated with the customers 204 and purchase histories of the customers 204. For example, profile information associated with a customer 204 may indicate that the customer 204 was much more likely to purchase items within a threshold amount of time after augmented reality elements including coupons associated with the items were sent to a display area of a client device 110 associated with the customer 204 than after augmented reality elements including other types of content were sent to a display area of a client device 110 associated with the customer 204. In the above example, based on the profile information, the experience enhancement engine 338 may be more likely to retrieve coupons from the content database 334 than other types of content to communicate to the interface engine 328, which may then generate an augmented reality element to be sent to a display area of a client device 110 associated with the customer 204 based on the content.

[0103] In some embodiments, the experience enhancement engine 338 also may use a machine-learned model to identify information to communicate to the interface engine

328. For example, the experience enhancement engine **338** may provide profile information associated with a customer **204** and information identifying an item as an input to a machine-learned model. In this example, the model may be trained using profile information associated with customers **204** stored in the customer database **314** (e.g., information describing augmented reality elements previously sent to display areas of client devices **110** associated with the customers **204**, preferences associated with the customers **204**, order/purchase histories associated with the customers **204**, etc.) and content stored in the content database **334**. Continuing with this example, the experience enhancement engine **338** may then receive an output from the model describing predicted likelihoods that the customer **204** will purchase the item after an augmented reality element that may be generated based on each content item associated with the item is sent to a display area of a client device **110** associated with the customer **204**. In the above example, the experience enhancement engine **338** may identify one or more content items to communicate to the interface engine **328** associated with the highest predicted likelihood(s).

[0104] The experience enhancement engine **338** also may match profile information associated with a customer **204** (e.g., stored in the customer database **314**) with one or more attributes of one or more items included among an inventory of a warehouse **150** (e.g., stored in the inventory database **304**). The experience enhancement engine **338** may then communicate information describing the match to the interface engine **328**, which may generate an augmented reality element based on the information. For example, if profile information stored in the customer database **314** indicates that a customer **204** is on a low-sugar diet, the experience enhancement engine **338** may match this information with attributes of items the customer **204** is holding corresponding to their sugar content and communicate information describing the match to the interface engine **328**. In this example, based on the information describing the match, an augmented reality element generated by the interface engine **328** comparing the items may highlight their sugar content. As an additional example, suppose that attributes of a roll of paper towels stored in the inventory database **304** indicate a brand, an item type, and a size associated with the paper towels and profile information associated with a customer **204** stored in the customer database **314** includes an order history that indicates the customer **204** orders the roll of paper towels with the corresponding attributes about once every month. In the above example, if the customer **204** has not ordered the roll of paper towels with the corresponding attributes in about a month, the experience enhancement engine **338** may match the attributes with the order history and communicate information describing the match to the interface engine **328**. Continuing with this example, based on the match, the interface engine **328** may then generate an augmented reality element that recommends the roll of paper towels (e.g., with a “Buy it again” heading) when overlaid onto a display area of a client device **110** associated with a shopper **208** fulfilling an order on behalf of the customer **204**. In this example, the shopper **208** may then communicate the recommendation to the customer **204**.

[0105] In various embodiments, the experience enhancement engine **338** also may match profile information associated with a customer **204** (e.g., stored in the customer database **314**) with attributes of other types of physical elements within a warehouse **150**, such as advertisements or

coupons (e.g., stored in the content database **334** and/or the inventory database **304**). In such embodiments, the experience enhancement engine **338** may communicate information describing the match to the interface engine **328**, which may generate an augmented reality element based on the information. For example, suppose that an advertisement for a can of coconut milk is detected within a display area of a client device **110** associated with a customer **204** and attributes of the advertisement include nutritional information associated with the can of coconut milk stored in the inventory database **304** corresponding to 13 grams of saturated fat. In this example, if a dietary restriction associated with the customer **204** stored in the customer database **314** indicates that the customer **204** is on a diet that prohibits saturated fat, the experience enhancement engine **338** may match the attribute with the dietary restriction and communicate information describing the match to the interface engine **328**. Continuing with this example, based on the information describing the match, the interface engine **328** may generate an augmented reality element that obscures the advertisement for the can of coconut milk when overlaid onto a portion of the display area of the client device **110** at which the advertisement is detected.

[0106] As described above, the experience enhancement engine **338** also may identify replacement items that may replace other items included among an inventory of a warehouse **150** (e.g., if the items that may be replaced are not available, if a customer **204** is likely to prefer the replacement items to the items that may be replaced, etc.). The experience enhancement engine **338** may do so upon receiving information identifying a set of potential replacement items from the item replacement engine **332**, in which the set of potential replacement items each have a set of attributes that have at least a threshold measure of similarity to a set of attributes of an item that may be replaced. The experience enhancement engine **338** may then identify one or more replacement items from the set of potential replacement items based on profile information associated with a customer **204** stored in the customer database **314** and attributes of the potential replacement items stored in the inventory database **304**. For example, if a set of preferences associated with a customer **204** indicates that the customer **204** prefers organic food items to non-organic food items, the experience enhancement engine **338** may be more likely to identify an organic replacement item than a non-organic replacement item. As an additional example, if a set of dietary restrictions associated with a customer **204** indicates that the customer **204** is a vegetarian, the experience enhancement engine **338** may be more likely to identify a vegetarian replacement item than a non-vegetarian replacement item. As yet another example, if an order history associated with a customer **204** indicates that the customer **204** has often ordered a potential replacement item, the experience enhancement engine **338** may be more likely to identify this potential replacement item as a replacement for an item than other potential replacement items. In embodiments in which a shopper **208** is fulfilling an order on behalf of a customer **204**, the experience enhancement engine **338** may retrieve profile information associated with the customer **204** based on details associated with the order identifying the customer **204** (e.g., stored in the transaction records database **308** or the training datasets **320**).

[0107] In some embodiments, the experience enhancement engine **338** may predict likelihoods that customers **204**

will accept potential replacement items and/or prefer potential replacement items to items that may be replaced. To do so, the experience enhancement engine 338 may access profile information associated with a customer 204 (e.g., stored in the customer database 314), attributes of each potential replacement item, and attributes of an item that may be replaced (e.g., stored in the inventory database 304). The experience enhancement engine 338 may then predict a likelihood that the customer 204 will accept each potential replacement item or prefer each potential replacement item to the item that may be replaced based on the profile information, the attributes of the item that may be replaced, and the attributes of each potential replacement item. In some embodiments, the experience enhancement engine 338 may predict the likelihoods by matching profile information associated with the customer 204 with a set of attributes of each potential replacement item and a set of attributes of the item that may be replaced. For example, suppose that a customer 204 is holding an item that they have never purchased and that the only profile information associated with the customer 204 that matches an attribute of the item corresponds to an item type for one or more items included in their purchase history. In this example, suppose also that the purchase history indicates that the customer 204 previously purchased a potential replacement item several times when it was on sale, such that the profile information matches multiple attributes of the potential replacement item (e.g., its brand, type, etc. and information indicating it is on sale). In the above example, the experience enhancement engine 338 may then predict a likelihood (e.g., 85%) that the customer 204 will prefer the potential replacement item to the item they are holding based on the match. Alternatively, if the item the customer 204 was holding in the above example is included in the customer's order or shopping list and the item is not available, the experience enhancement engine 338 may predict a likelihood (e.g., 85%) that the customer 204 will accept the potential replacement item in an analogous manner.

[0108] The experience enhancement engine 338 may then identify replacement items from potential replacement items based on the predicted likelihoods that customers 204 will accept the replacement items and/or prefer the replacement items to items that may be replaced. In some embodiments, the experience enhancement engine 338 may identify one or more replacement items associated with at least a threshold likelihood from the set of potential replacement items. Alternatively, the experience enhancement engine 338 may rank the set of potential replacement items based on the predicted likelihood associated with each potential replacement item and identify one or more replacement items based on the ranking (e.g., one or more replacement items associated with at least a threshold ranking). In embodiments in which the experience enhancement engine 338 identifies one or more replacement items that may replace another item that is not available, the experience enhancement engine 338 may then communicate information to the interface engine 328 identifying the replacement item(s). In embodiments in which the experience enhancement engine 338 identifies one or more replacement items that a customer 204 is likely to prefer to another item that may be replaced, the experience enhancement engine 338 may then communicate information to the interface engine 328 identifying the replacement item(s) and information indicating why the customer 204 is likely to prefer the replacement item(s) to the item that may

be replaced. The interface engine 328 may then generate an augmented reality element that recommends the replacement item(s). In embodiments in which multiple replacement items are identified, an augmented reality element that recommends the replacement items may list the replacement items in an order based on the ranking described above.

[0109] In some embodiments, the experience enhancement engine 338 also may identify replacement items using a machine-learned model. For example, the experience enhancement engine 338 may provide attributes of a customer 204 as an input to a machine-learned model that is trained using profile information associated with customers 204 (e.g., order/purchase histories, preferences, etc.) stored in the customer database 314 and attributes of items included among an inventory of a warehouse 150 stored in the inventory database 304. The experience enhancement engine 338 may then receive an output from the model describing predicted likelihoods that the customer 204 will purchase each item included among the inventory of the warehouse 150. In this example, if the experience enhancement engine 338 is identifying a replacement item that replaces an item that is not available, the experience enhancement engine 338 may identify a potential replacement item as a replacement item if the potential replacement item is associated with at least a threshold likelihood. Alternatively, in the above example, if the experience enhancement engine 338 is identifying a replacement item that replaces an item that is available, the experience enhancement engine 338 may identify a potential replacement item as a replacement item if the potential replacement item is associated with a likelihood that is greater than a likelihood associated with the item that may be replaced.

[0110] FIG. 3B is a diagram of a wayfinding application 118, according to one embodiment. In various embodiments, the wayfinding application 118 may include different or additional modules than those described in conjunction with FIG. 3B. Furthermore, in some embodiments, the wayfinding application 118 includes fewer modules than those described in conjunction with FIG. 3B.

[0111] The components of the wayfinding application 118 illustrated in FIG. 3B function in the same manner described above in conjunction with FIG. 3A. As described above, the wayfinding application 118 displays augmented reality elements for navigating to a location of an item within a warehouse 150. As also described above, the wayfinding application 118 may be executed on a client device 110 associated with a shopper 208 fulfilling an order on behalf of a customer 204 or a client device 110 associated with a customer 204 of a retailer associated with a warehouse 150, in which the customer 204 may or may not be associated with the online concierge system 102. Furthermore, in various embodiments, one or more components of the wayfinding application 118 may interact with one or more components of the online concierge system 102, the CMA 206, and/or the SMA 212, while in other embodiments, the wayfinding application 118 is a standalone application.

Customer Mobile Application

[0112] FIG. 4A is a diagram of the customer mobile application (CMA) 206, according to one or more embodiments. The CMA 206 includes an ordering interface 402, which provides an interactive interface which a customer 204 may use to browse through and select products and place an order. The CMA 206 also includes a system

communication interface **404** which, among other functions, receives inventory information from the online concierge system **102** and transmits order information to the system **102**. The CMA **206** also includes a preferences management interface **406** which allows a customer **204** to manage basic information associated with his/her account, such as his/her home address and payment instruments. The preferences management interface **406** may also allow a customer **204** to manage other details such as his/her favorite or preferred warehouses **150**, preferred delivery times, special instructions for delivery, and so on.

Shopper Mobile Application

[0113] FIG. 4B is a diagram of the shopper mobile application (SMA) **212**, according to one or more embodiments. The SMA **212** includes a barcode scanning module **420**, which allows a shopper **208** to scan an item at a warehouse **150** (such as a can of soup on the shelf at a grocery store). The barcode scanning module **420** may also include an interface, which allows a shopper **208** to manually enter information describing an item (such as its serial number, SKU, quantity and/or weight) if a barcode is not available to be scanned. The SMA **212** also includes a basket manager **422**, which maintains a running record of items collected by a shopper **208** for purchase at a warehouse **150**. This running record of items is commonly known as a “basket.” In one embodiment, the barcode scanning module **420** transmits information describing each item (such as its cost, quantity, weight, etc.) to the basket manager **422**, which updates its basket accordingly. The SMA **212** also includes a system communication interface **424**, which interacts with the online concierge system **102**. For example, the system communication interface **424** receives an order from the online concierge system **102** and transmits the contents of a basket of items to the online concierge system **102**. The SMA **212** also includes an image encoder **426** which encodes the contents of a basket into an image. For example, the image encoder **426** may encode a basket of goods (with an identification of each item) into a QR code, which may then be scanned by an employee of a warehouse **150** at check-out.

Displaying Augmented Reality Elements for Navigating to a Location of an Item within a Warehouse

[0114] FIG. 5 is a flowchart of a method for displaying augmented reality elements for navigating to a location of an item within a warehouse **150**, according to one or more embodiments. In various embodiments, the method includes different or additional steps than those described in conjunction with FIG. 5. Further, in some embodiments, the steps of the method may be performed in different orders than the order described in conjunction with FIG. 5. The method described in conjunction with FIG. 5 may be carried out by the wayfinding application **118** and/or the online concierge system **102** in various embodiments, while in other embodiments, the steps of the method are performed by any application and/or any online system capable of retrieving items.

[0115] In various embodiments, the user of the client device **110** described below is a shopper **208** associated with the online concierge system **102** fulfilling an order on behalf of a customer **204**, while in other embodiments, the user is a customer **204** of the retailer associated with the warehouse **150**, who may or may not be associated with the online concierge system **102**. Therefore, in some embodiments,

“customer **204**” may refer to a user of the online concierge system **102** who has placed an order to be fulfilled by a shopper **208**, while in other embodiments, “customer **204**” may refer to the user of the client device **110** who is doing their own shopping at the warehouse **150**.

[0116] The wayfinding application **118** or the online concierge system **102** may receive **505** (e.g., via the route engine **324**) a current location of the client device **110** (e.g., an augmented reality device or a mobile device) within a warehouse **150** (e.g., a physical grocery store), in which the warehouse **150** is associated with a retailer and multiple items included among an inventory of the warehouse **150**. In some embodiments, the location of the client device **110** within the warehouse **150** may be received **505** from a location tracking system **152** at the warehouse **150**. For example, the location tracking system **152** at the warehouse **150** may include one or more sensors **156** that detect client devices **110** scanning for Wi-Fi networks and record MAC addresses associated with any detected client devices **110**. In this example, the location tracking system **152** may track the location of the client device **110** (e.g., in various aisles or departments) as it moves throughout the warehouse **150** based on its MAC address and image or video data of the user of the client device **110** captured by one or more cameras included in a surveillance system **154** of the location tracking system **152**. Continuing with this example, the wayfinding application **118** or the online concierge system **102** may receive **505** the location of the client device **110** within the warehouse **150** from the location tracking system **152**. In some embodiments, the wayfinding application **118** or the online concierge system **102** also or alternatively may receive other types of information from the location tracking system **152** (e.g., information identifying the warehouse **150**, a time at which the client device **110** was at the warehouse **150** or at a specific location within the warehouse **150**, etc.). In the above example, the wayfinding application **118** or the online concierge system **102** may receive **505** the location of the client device **110** within the warehouse **150** in association with a warehouse identifier associated with the warehouse **150** and a timestamp indicating a date and time at which the client device **110** was at the location within the warehouse **150**.

[0117] The wayfinding application **118** or the online concierge system **102** also may receive information identifying the warehouse **150**. Information identifying the warehouse **150** may be received from a location tracking system **152** at the warehouse **150**, as described above. The wayfinding application **118** or the online concierge system **102** also or alternatively may receive (e.g., using the route engine **324**) information identifying the warehouse **150** from the client device **110** via GPS triangulation, cell tower triangulation, Wi-Fi positioning, or any other suitable technique. For example, if the client device **110** is at the warehouse **150**, the wayfinding application **118** or the online concierge system **102** may track the location of the client device **110** (e.g., via GPS triangulation), access information describing the locations of various warehouses **150** (e.g., stored in the layout database **322** or a warehouse database), and identify the warehouse **150** based on the location of the client device **110**. In embodiments in which the user of the client device **110** is a shopper **208** fulfilling an order on behalf of the customer **204**, the online concierge system **102** may identify the warehouse **150** at which the order is to be fulfilled (e.g., using the shopper management engine **310**). In such

embodiments, the online concierge system 102 may then communicate information identifying the warehouse 150 to the wayfinding application 118.

[0118] In some embodiments, the wayfinding application 118 or the online concierge system 102 also or alternatively may receive information identifying the warehouse 150 via an interface that the wayfinding application 118 or the online concierge system 102 generates and transmits (e.g., using the interface engine 328). In such embodiments, the interface may include one or more fields, buttons, or other interactive elements for receiving various inputs from the user of the client device 110. For example, if the user of the client device 110 is at the warehouse 150, the user may select an option to share the location of the client device 110 by clicking on a button included in the interface. Alternatively, in the above example, the user may select the warehouse 150 from a drop-down menu included in the interface or type a name of the retailer associated with the warehouse 150 into a search box included in the interface and select the warehouse 150 from a list of results.

[0119] In various embodiments, the wayfinding application 118 or the online concierge system 102 also may receive information identifying one or more items (“target items”) included among the inventory of the warehouse 150 to be collected from the warehouse 150 by the user of the client device 110. The target item(s) may correspond to one or more items included in an order or a shopping list. In some embodiments, the wayfinding application 118 or the online concierge system 102 may receive information identifying the target item(s) in an input from the user of the client device 110 via the interface described above. In such embodiments, upon receiving the input from the user via the interface, the wayfinding application 118 or the online concierge system 102 may retrieve information identifying items included among the inventory of the warehouse 150 based on the input (e.g., items having attributes with values that match at least a portion of the input identified by leveraging item embeddings). Via the interface, one or more items may then be added to a list of target items (e.g., a shopping list) or subsequently removed from the list. For example, the interface may include a search box that allows the user to type in a name of an item, a description of an item, etc. In this example, the user may add one or more items included in a list of results to a list of target items and search for additional items to add to the list of target items. In embodiments in which the user of the client device 110 is a shopper 208 fulfilling an order on behalf of the customer 204, the online concierge system 102 may receive information identifying the target item(s) included in the order (e.g., via the order fulfillment engine 306). In such embodiments, the online concierge system 102 may then communicate information identifying the target item(s) to the wayfinding application 118.

[0120] The wayfinding application 118 or the online concierge system 102 then accesses 510 (e.g., using the route engine 324) a layout of the warehouse 150 (e.g., from the layout database 322). In embodiments in which layouts of multiple warehouses 150 are maintained (e.g., in the layout database 322), the layout of the warehouse 150 accessed 510 by the wayfinding application 118 or the online concierge system 102 may be identified based on information identifying the warehouse 150 received by the methods described above. In some embodiments, the layout of the warehouse 150 may be received from the retailer associated with the

warehouse 150 and updated by the retailer or the wayfinding application 118 or the online concierge system 102 (e.g., using the item/location detection engine 330), as described below. The layout of the warehouse 150 may be stored in association with information that uniquely identifies the warehouse 150, such as a name of the warehouse 150, a geographic location of the warehouse 150, etc. For example, the layout of the warehouse 150 may be stored in association with GPS coordinates associated with the warehouse 150, a combination of a name of the retailer associated with the warehouse 150 and a city and state in which the warehouse 150 is located, etc.

[0121] The layout of the warehouse 150 may describe departments and/or physical elements within the warehouse 150, such as organizational elements, including aisles, shelves within the aisles, various display areas (e.g., frozen or refrigerated display cases, display tables, and sample and promotional displays), etc., as well as their arrangement within the warehouse 150. In addition to organizational elements, the layout of the warehouse 150 also may describe additional physical elements within the warehouse 150, such as items included among the inventory of the warehouse 150, service counters (e.g., for various departments and/or checkout counters), and various building elements (e.g., entrances/exits, floors, ceilings, walls, stairs, elevators, etc.), and their arrangement within the warehouse 150. The layout of the warehouse 150 also may include one or more images of departments and physical elements within the warehouse 150 captured from one or more angles, positions, etc. In some embodiments, the layout of the warehouse 150 also may describe dimensions associated with the warehouse 150 (e.g., dimensions associated with departments and/or organizational elements within the warehouse 150). For example, the layout of the warehouse 150 may describe lengths of various aisles, distances between the aisles, heights of shelves, etc. Furthermore, in various embodiments, the layout of the warehouse 150 may describe congestion/traffic within the warehouse 150. For example, the layout of the warehouse 150 may describe certain times or days of the week when particular service counters or aisles with sample displays are more congested than others.

[0122] Additionally, the layout of the warehouse 150 may describe a set of locations within the warehouse 150 associated with each item included among its inventory. A location within the warehouse 150 associated with an item may correspond to a particular area within the warehouse 150 at which the item may be collected. For example, the layout of the warehouse 150 may describe one or more locations within the warehouse 150 associated with an item, in which each location corresponds to a particular aisle, a particular shelf, a particular display table, a particular promotional display, etc., at which the item may be collected.

[0123] In various embodiments, once the layout of the warehouse 150 has been accessed 510, the wayfinding application 118 or the online concierge system 102 also may detect (e.g., using the item/location detection engine 330) the location and/or orientation of the client device 110 within the warehouse 150. In such embodiments, the wayfinding application 118 or the online concierge system 102 may receive (e.g., via the interface engine 328) image or video data including a display area of the client device 110 captured by a camera of the client device 110 and detect the location and/or orientation of the client device 110 based on the image or video data and the layout of the warehouse 150.

For example, if the client device **110** is a smartphone, the camera of the smartphone may capture image or video data that is visible within the display area of the smartphone to the user of the smartphone. Alternatively, in the above example, if the client device **110** is a pair of augmented reality glasses, the camera of the augmented reality glasses may capture image or video data that is visible to the user wearing the augmented reality glasses. In the above example, the image or video data may depict various physical elements within the warehouse **150**, such as items included among the inventory of the warehouse **150**, service counters (e.g., for various departments and/or checkout counters), organizational elements (e.g., aisles, shelves, display areas, etc.), and various building elements (e.g., entrances/exits, floors, ceilings, walls, stairs, elevators, etc.). Continuing with this example, the wayfinding application **118** or the online concierge system **102** may then compare the physical elements included in the image or video data to those of various locations within the warehouse **150** included in the layout of the warehouse **150**. In this example, the wayfinding application **118** or the online concierge system **102** may then detect a location and/or orientation within the warehouse **150** corresponding to the location and/or orientation of the client device **110** if the physical elements included in the image or video data have at least a threshold measure of similarity to the physical elements at the location.

[0124] The wayfinding application **118** or the online concierge system **102** then identifies **515** (e.g., using the route engine **324**) a route for navigating from the location of the client device **110** within the warehouse **150** to one or more locations within the warehouse **150** associated with the target item(s). The wayfinding application **118** or the online concierge system **102** may identify **515** the route based on the location of the client device **110** within the warehouse **150**, information identifying the target item(s), and the layout of the warehouse **150**. For example, as shown in FIG. 6, which illustrates a route for navigating from a location of a client device **110** within a warehouse **150** to one or more locations within the warehouse **150** associated with one or more items, according to one or more embodiments, suppose that the client device **110** is at a location within the warehouse **150** corresponding to an entrance **605a** or exit **605b** of the warehouse **150**. In this example, based on the layout of the warehouse **150**, the wayfinding application **118** or the online concierge system **102** may identify **515** a route **610** for navigating from the entrance **605a** to the locations of six target items within the warehouse **150**, in which the six target items are included in a shopping list, and then to the exit **605ab**. In the above example, the route **610** may indicate an order in which the target items may be collected and the locations along the route **610** (e.g., specific shelves or locations within specific display cases) at which the target items may be collected.

[0125] The wayfinding application **118** or the online concierge system **102** may identify **515** the route **610** using various techniques. In some embodiments, the route **610** may be identified **515** based on a distance from the location of the client device **110** within the warehouse **150** to the location(s) within the warehouse **150** associated with the target item(s). For example, the wayfinding application **118** or the online concierge system **102** may identify **515** the route **610** that requires the user of the client device **110** to travel a minimum distance to collect the target item(s). In

various embodiments, the wayfinding application **118** or the online concierge system **102** also or alternatively may identify **515** the route **610** based on a predicted amount of time required to travel from the location of the client device **110** within the warehouse **150** to the location(s) within the warehouse **150** associated with the target item(s). For example, based on information describing congestion/traffic within the warehouse **150** during different times of the day described in the layout of the warehouse **150**, the wayfinding application **118** or the online concierge system **102** may identify **515** the route **610** that is predicted to take the user of the client device **110** a shortest amount of time to travel to collect the target item(s).

[0126] In some embodiments, the wayfinding application **118** or the online concierge system **102** also may identify **515** the route **610** using a machine-learned model. For example, the wayfinding application **118** or the online concierge system **102** may identify **515** the route **610** using a machine-learned model that is trained based on information describing routes **610** and other information associated with the routes **610** (e.g., stored in the route database **326**). In this example, the wayfinding application **118** or the online concierge system **102** may provide an input to the machine-learned model including the location of the client device **110** within the warehouse **150**, information identifying the target item(s), and the layout of the warehouse **150**. Continuing with this example, the wayfinding application **118** or the online concierge system **102** may receive an output from the model identifying **515** the route **610** that requires the user of the client device **110** to travel a minimum distance to collect the target item(s) and/or the route **610** that is predicted to take the user a shortest amount of time to collect the target item(s).

[0127] In some embodiments, the wayfinding application **118** or the online concierge system **102** may be configured to store information describing the route **610** (e.g., in the route database **326**). In such embodiments, the information describing the route **610** may be stored in association with various types of information. The information describing the route **610** may be stored in association with information describing the client device **110** associated with the route **610** (e.g., an IP address associated with the client device **110**), the user associated with the client device **110** (e.g., a username, email address, or other identifier associated with the user), and/or the warehouse **150** associated with the route **610** (e.g., a warehouse identifier). The information describing the route **610** also or alternatively may be stored in association with a time associated with the route **610** (e.g., a timestamp indicating when it was identified **515**), information identifying the target item(s) to be collected along the route **610**, a distance required to travel the route **610**, an amount of time (predicted or actual) required to travel the route **610**, or any other suitable types of information that may be associated with the route **610**.

[0128] Referring back to FIG. 5, the wayfinding application **118** or the online concierge system **102** then generates **520** (e.g., using the interface engine **328**) a set of augmented reality elements that includes a set of instructions for navigating the route **610** within the warehouse **150**. The set of augmented reality elements may include turn-by-turn instructions for navigating the route **610**. For example, based on the route **610**, the wayfinding application **118** or the online concierge system **102** may generate **520** the set of augmented reality elements that correspond to signs that,

when overlaid onto a portion of the display area of the client device 110, provide instructions for navigating the route 610.

[0129] The wayfinding application 118 or the online concierge system 102 then sends 525 (e.g., using the interface engine 328) the set of augmented reality elements to the display area of the client device 110. An augmented reality element may be overlaid onto a portion of the display area of the client device 110 at which an item is detected (e.g., by the item/location detection engine 330) or onto another portion of the display area, such that it appears to float in the portion of the display area or near an item detected within the display area. For example, FIG. 7A illustrates a set of augmented reality elements 715a for navigating a route 610 within a warehouse 150 to one or more locations associated with one or more items, according to one or more embodiments, and continues the example described above in conjunction with FIG. 6. In this example, the set of augmented reality elements 715a correspond to arrows that are overlaid onto a portion of the display area of the client device 110 at which the floor of the warehouse 150 is detected, such that the arrows provide instructions for navigating the route 610. In this example, the user of the client device 110 may follow the arrows while at the warehouse 150 to navigate to locations associated with the six target items.

[0130] In some embodiments, the wayfinding application 118 or the online concierge system 102 also may generate (e.g., using the interface engine 328) an augmented reality element 715 that includes a list of the target item(s) and send (e.g., using the interface engine 328) the augmented reality element 715 to the display area of the client device 110. As described above, the target item(s) may be identified by the customer 204 (e.g., in an order received from the customer 204) or by the user of the client device 110 (e.g., in an input received via the interface described above). For example, if the customer 204 places a delivery order via the online concierge system 102, the wayfinding application 118 or the online concierge system 102 may generate an augmented reality element 715 that includes a list of the target item(s), in which each target item corresponds to an item included in the order. In this example, the augmented reality element 715 subsequently may be sent to the display area of the client device 110, in which the user of the client device 110 is a shopper 208 assigned to fulfill the order. As an additional example, if information identifying the target item(s) is received in an input from the user of the client device 110 via the interface described above, in which each target item is included in a shopping list, the wayfinding application 118 or the online concierge system 102 may generate an augmented reality element 715 that includes the shopping list. In some embodiments, a list of the target item(s) included in an augmented reality element 715 may be ordered and/or numbered based on the route 610. In the example of FIG. 7A, an augmented reality element 715b including a list of the target items is overlaid onto an upper-left corner of the display area of the client device 110, in which the list of the six target items corresponds to a shopping list. In this example, the target items in the shopping list included in the augmented reality element 715b are ordered and numbered based on an order in which they will be encountered along the route 610.

[0131] In various embodiments, the wayfinding application 118 or the online concierge system 102 also may generate (e.g., using the interface engine 328) an augmented

reality element 715 that includes the route 610 for navigating from the location of the client device 110 within the warehouse 150 to the location(s) within the warehouse 150 associated with the target item(s). The augmented reality element 715 may indicate the location of the client device 110 within the warehouse 150, the location(s) within the warehouse 150 along the route 610 at which the target item(s) may be collected, and an order in which the target item(s) may be collected. For example, as shown in FIG. 7A, an augmented reality element 715c that includes the route 610 is overlaid onto an upper-right corner of the display area of the client device 110. In this example, the augmented reality element 715c depicts the layout of the warehouse 150, in which the location of the client device 110 and locations associated with the six target items included in the shopping list included in augmented reality element 715b are represented by icons that are connected by a line representing the route 610. In this example, the locations associated with the target items are numbered based on an order in which the user of the client device 110 will encounter the locations associated with the target items by following the route 610.

[0132] Referring back to FIG. 5, the wayfinding application 118 or the online concierge system 102 then detects 530 (e.g., via the item/location detection engine 330), within the display area of the client device 110, a location within the warehouse 150 associated with a target item. For example, suppose that the target item is associated with a particular shelf within the warehouse 150 at which the target item may be collected. In this example, the wayfinding application 118 or the online concierge system 102 may receive image or video data captured by the camera of the client device 110 and compare physical elements included in the image or video data to those at the location associated with the target item included in the layout of the warehouse 150. Continuing with this example, the wayfinding application 118 or the online concierge system 102 may detect 530 the location within the warehouse 150 associated with the target item if the physical elements included in the image or video data have at least a threshold measure of similarity to the physical elements (i.e., the particular shelf and any surrounding physical elements) at the location associated with the target item.

[0133] In some embodiments, the wayfinding application 118 or the online concierge system 102 may generate (e.g., using the interface engine 328) an augmented reality element 715 that calls attention to the location within the warehouse 150 associated with the target item and send (e.g., using the interface engine 328) the augmented reality element 715 to the display area of the client device 110. The augmented reality element 715 may call attention to the location within the warehouse 150 associated with the target item by highlighting it (e.g., with one or more colors) or with icons (e.g., arrows or circles), animations, etc. An example of the augmented reality element 715 is shown in FIG. 7B, which illustrates a set of augmented reality elements 715 for navigating a route 610 within a warehouse 150 to one or more locations associated with one or more items, according to one or more embodiments, and continues the example described above in conjunction with FIGS. 6 and 7A. In the example of FIG. 7B, by following the augmented reality elements 715a for navigating the route 610, the user of the client device 110 may approach the location within the warehouse 150 associated with a target item. In this

example, an augmented reality element **715d** that calls attention to the location of the target item detected **530** within the display area of the client device **110** is an icon with a number (1) that corresponds to the number of the target item within the shopping list included in augmented reality element **715b**. The same icon also represents the location associated with the target item included in augmented reality element **715c** that includes the route **610**. Continuing with this example, the augmented reality element **715d** that calls attention to the location associated with the target item hovers next to and points to a particular portion of the display area in which a shelf corresponding to the location associated with the target item is detected **530**.

[0134] In various embodiments, as physical elements, locations, etc. detected **530** within the display area of the client device **110** (e.g., by the item/location detection engine **330**) move within the display area, one or more augmented reality elements **715** may move as well. For example, in FIG. 7B, if the location associated with the target item moves within the display area of the client device **110**, the augmented reality element **715d** may move as well, such that it tracks the movement of the location. In some embodiments, augmented reality elements **715** may be hidden by the user of the client device **110**, such that they are not visible within the display area of the client device **110**. For example, the user of the client device **110** may switch between different modes via the interface described above, such that in one mode, all types of augmented reality elements **715** are visible within the display area of the client device **110**, in another mode, only certain types of augmented reality elements **715** (e.g., those including shopping lists and routes **610**) are visible, and in yet another mode, none of the types of augmented reality elements **715** are visible.

[0135] Referring again to FIG. 5, once the wayfinding application **118** or the online concierge system **102** has detected **530** the location within the warehouse **150** associated with the target item, the wayfinding application **118** or the online concierge system **102** may determine **535** (e.g., using the item/location detection engine **330**) whether the target item is at the location. In some embodiments, the wayfinding application **118** or the online concierge system **102** may do so based on one or more images of the target item (e.g., one or more images included in a catalog stored in the inventory database **304**) and image or video data captured by the camera of the client device **110**. For example, the wayfinding application **118** or the online concierge system **102** may receive image or video data captured by the camera of the client device **110**, which includes various physical elements displayed within the display area of the client device **110**. In this example, once the wayfinding application **118** or the online concierge system **102** detects **530** the location within the warehouse **150** associated with the target item, it may access images of the target item included in a catalog (e.g., stored in the inventory database **304**) and compare the physical elements to the images of the target item. Continuing with this example, if the physical elements have at least a threshold measure of similarity to the images of the target item included in the catalog, the wayfinding application **118** or the online concierge system **102** may determine **535** the target item is at the location associated with the target item. However, in the above example, if the physical elements do not have at least the threshold measure of similarity to the images of the target

item included in the catalog, the wayfinding application **118** or the online concierge system **102** may determine **535** the target item is not at the location associated with the target item.

[0136] In some embodiments, the wayfinding application **118** or the online concierge system **102** also may determine **535** whether the target item is at the location within the warehouse **150** associated with the target item using a machine-learned model. For example, the wayfinding application **118** or the online concierge system **102** may provide image or video data captured by the camera of the client device **110** as an input to a machine-learned model that is trained using images of various physical elements within the warehouse **150**, such as the target item and other items included among the inventory of the warehouse **150**. The wayfinding application **118** or the online concierge system **102** may then receive an output from the model describing predicted likelihoods that each physical element is detected in the image or video data and determine **535** that the target item is detected if the corresponding predicted likelihood is at least a threshold likelihood. Alternatively, the wayfinding application **118** or the online concierge system **102** may determine **535** that the target item is not detected if the corresponding predicted likelihood is less than the threshold likelihood.

[0137] In some embodiments, if the wayfinding application **118** or the online concierge system **102** determines **535** that the target item is not at the location within the warehouse **150** associated with the target item, it also may determine (e.g., using the item/location detection engine **330**) whether the target item is detected elsewhere within the warehouse **150**. In such embodiments, this determination may be made within a threshold amount of time of detecting **530** the location associated with the target item, within a threshold distance of detecting **530** the location associated with the target item, etc. For example, suppose that the target item is misplaced on a shelf above or below a specific shelf corresponding to the location within the warehouse **150** associated with the target item. In this example, once the wayfinding application **118** or the online concierge system **102** has detected **530** the location within the warehouse **150** associated with the target item and determines **535** that the target item is not at the location, the wayfinding application **118** or the online concierge system **102** may determine whether the target item is detected within the display area of the client device **110** while the location is also detected **530** within the display area. If the wayfinding application **118** or the online concierge system **102** determines **535** that the target item is not at the location within the warehouse **150** associated with the target item (or elsewhere within the warehouse **150**, in some embodiments), inventory information associated with the warehouse **150** (e.g., stored in the inventory database **304**) may be updated (e.g., using the inventory management engine **302**) to reflect that the target item is out of stock at the warehouse **150**.

[0138] Upon determining **535** that the target item is not at the location within the warehouse **150** associated with the target item (or elsewhere within the warehouse **150**, in various embodiments), the wayfinding application **118** or the online concierge system **102** may receive information identifying a replacement item included among the inventory of the warehouse **150**. In embodiments in which the user of the client device **110** is a shopper **208** fulfilling a delivery order on behalf of the customer **204**, the replacement item may be

identified by the customer 204. For example, suppose that the target item is a particular brand and flavor of ice cream that is not available and the user of the client device 110 is a shopper 208 fulfilling an order on behalf of the customer 204. In this example, the user may identify potential replacement items that have at least a threshold measure of similarity to the target item (e.g., other flavors of ice cream of the same brand or other brands of ice cream of the same flavor). In the above example, the user may then communicate the potential replacement items to a client device 110 associated with the customer 204 (e.g., by capturing an image or a video of them and sending the image or video via text or email, by listing or describing them during a phone call or in a text message, etc.). Continuing with this example, the user may then receive a response from the customer 204 identifying a replacement item included among the potential replacement items.

[0139] Alternatively, in some embodiments, the wayfinding application 118 or the online concierge system 102 may identify (e.g., using the item replacement engine 332) the replacement item. In such embodiments, the wayfinding application 118 or the online concierge system 102 may do so by accessing a set of attributes of the target item and attributes of other items included among the inventory of the warehouse 150 (e.g., stored in the inventory database 304) and identifying the replacement item that has a set of attributes that have at least a threshold measure of similarity to the set of attributes of the target item. In various embodiments, the wayfinding application 118 or the online concierge system 102 may identify the replacement item by accessing a taxonomy (e.g., maintained by the inventory management engine 302) and identifying a set of potential replacement items included in a most specific category in which the target item is included. The wayfinding application 118 or the online concierge system 102 may then compare a set of attributes of each potential replacement item with a set of attributes of the target item and identify the replacement item from the set of potential replacement items based on the comparison, in which the set of attributes of the replacement item has at least a threshold measure of similarity to the set of attributes of the target item.

[0140] In various embodiments, the wayfinding application 118 or the online concierge system 102 also may identify (e.g., using the experience enhancement engine 338) the replacement item from a set of potential replacement items that each have a set of attributes that have at least a threshold measure of similarity to a set of attributes of the target item. In such embodiments, the replacement item may be identified based on profile information associated with the customer 204 (e.g., stored in the customer database 314) and attributes of the potential replacement items (e.g., stored in the inventory database 304). The wayfinding application 118 or the online concierge system 102 may identify (e.g., using the item replacement engine 332) the set of potential replacement items by accessing attributes of the target item and attributes of other items included among the inventory of the warehouse 150 or by accessing a taxonomy (e.g., maintained by the inventory management engine 302), as described above.

[0141] Once the replacement item is identified, the wayfinding application 118 or the online concierge system 102 may then alert 540 the user of the client device 110 to the replacement item by generating 545 (e.g., using the interface engine 328) an augmented reality element 715 that calls

attention to the replacement item and sending 550 (e.g., using the interface engine 328) the augmented reality element 715 to the display area of the client device 110. Similar to the augmented reality element 715 described above, the augmented reality element 715 may call attention to the replacement item by highlighting it (e.g., with one or more colors) or with icons (e.g., arrows or circles), animations, etc. In some embodiments, prior to generating 545 the augmented reality element 715 that calls attention to the replacement item, the wayfinding application 118 or the online concierge system 102 may first detect (e.g., via the item/location detection engine 330) the location within the warehouse 150 associated with the replacement item. In such embodiments, the wayfinding application 118 or the online concierge system 102 may then generate (e.g., using the interface engine 328) an augmented reality element 715 that calls attention to the location within the warehouse 150 associated with the replacement item and send (e.g., using the interface engine 328) the augmented reality element 715 to the display area of the client device 110. Furthermore, in embodiments in which the wayfinding application 118 or the online concierge system 102 generates and sends an augmented reality element 715 that calls attention to the location within the warehouse 150 associated with the target item, the wayfinding application 118 or the online concierge system 102 may update (e.g., using the interface engine 328) the augmented reality element 715 and send the updated augmented reality element 715 to the display area of the client device 110.

[0142] An example of the augmented reality element 715 that calls attention to the replacement item is shown in FIG. 7C, which illustrates a set of augmented reality elements 715 that calls attention to a replacement item within a warehouse 150, according to one or more embodiments, and continues the example described above in conjunction with FIGS. 6, 7A, and 7B. In the example of FIG. 7C, once the user of the client device 110 arrives at the location within the warehouse 150 associated with the target item, the wayfinding application 118 or the online concierge system 102 determines 535 that the target item is not at the location and detects the location within the warehouse 150 associated with the replacement item. The wayfinding application 118 or the online concierge system 102 then updates the augmented reality element 715d that calls attention to the location within the warehouse 150 associated with the target item by graying it out and sends the updated augmented reality element 715d to the display area of the client device 110. The wayfinding application 118 or the online concierge system 102 also generates an augmented reality element 715e that calls attention to the location of the replacement item detected within the display area of the client device 110, which is an icon with a number (1A) that corresponds to the number of the replacement item within the shopping list included in augmented reality element 715b. In this example, once sent to the display area of the client device 110, the augmented reality element 715e hovers next to a particular portion of the display area in which a shelf corresponding to the location associated with the replacement item is detected (e.g., by the item/location detection engine 330). In the above example, the wayfinding application 118 or the online concierge system 102 alerts 540 the user to the replacement item by generating 545 an additional augmented reality element 715f that calls attention to the replacement item and sending 550 the additional augmented

reality element **715f** to the display area of the client device **110**. Continuing with this example, the additional augmented reality element **715f** that calls attention to the replacement item is a halo that is overlaid onto a portion of the display area of the client device **110** at which the replacement item is detected (e.g., by the item/location detection engine **330**).

[0143] In some embodiments, the wayfinding application **118** or the online concierge system **102** also may generate (e.g., via the interface engine **328**) an augmented reality element **715** that recommends the replacement item and send (e.g., via the interface engine **328**) the augmented reality element **715** for display to the user of the client device **110**. In such embodiments, the augmented reality element **715** recommending the replacement item may include information associated with the replacement item (e.g., brand name, item type, size, nutritional information, price, discounts or coupons that may apply to the replacement item, etc.). Furthermore, in such embodiments, the augmented reality element **715** recommending the replacement item also may indicate a reason that the replacement item is being recommended. As shown in the example of FIG. 7C, an augmented reality element **715g** indicates that the replacement item is being recommended because the target item is out of stock and includes information associated with the replacement item (i.e., the brand name and item type). In some embodiments, the user of the client device **110** may adjust one or more settings to change augmented reality elements **715** that recommend replacement items, such as types of information about the replacement items that are included in the augmented reality element **715** (e.g., unit price, item price, brand, etc.), criteria for recommending replacement items, etc. via the interface described above.

[0144] In some embodiments, if the wayfinding application **118** or the online concierge system **102** generates an augmented reality element **715** that includes a list of the target items, the wayfinding application **118** or the online concierge system **102** may update (e.g., using the interface engine **328**) the augmented reality element **715**. In such embodiments, the wayfinding application **118** or the online concierge system **102** may do so if a target item or a replacement item is unavailable, if a target item or a replacement item is collected, if a replacement item is identified, etc. For example, as shown in FIG. 7C, once the wayfinding application **118** or the online concierge system **102** determines **535** that the target item (i.e., Brand A refried beans) is not at the location within the warehouse **150** associated with the target item, it may update the shopping list included in augmented reality element **715b** so that the target item is grayed out and includes a note indicating that the target item is not in stock. Furthermore, in the above example, once the replacement item (i.e., Brand E refried beans) is identified, the augmented reality element **715b** is updated so that the shopping list includes the replacement item and a note indicating it is a replacement item. In the above example, the replacement item may not be added to the augmented reality element **715b** until an input is received from the user of the client device **110** (e.g., via the interface described above) accepting the replacement item.

[0145] In some embodiments, once the replacement item is identified, the wayfinding application **118** or the online concierge system **102** may modify (e.g., using the route engine **324**) the route **610** based on a location within the

warehouse **150** associated with the replacement item (e.g., if the replacement item is not located near the target item). For example, once the replacement item is identified, the wayfinding application **118** or the online concierge system **102** may identify a location within the warehouse **150** associated with the replacement item based on the layout of the warehouse **150**. In this example, the wayfinding application **118** or the online concierge system **102** may then modify the route **610** such that the modified route **610** is for navigating from the location associated with the target item to the location associated with the replacement item and the locations associated with any remaining target items. In embodiments in which the wayfinding application **118** or the online concierge system **102** modifies the route **610**, the route **610** may be modified using the same techniques described above for identifying **515** the route **610** and information describing the modified route **610** may be stored (e.g., in the route database **326** in association with various types of information, as described above). In various embodiments, the route **610** may not be modified until an input is received from the user of the client device **110** (e.g., via the interface described above) accepting the replacement item. Furthermore, in embodiments in which the route **610** is modified, the wayfinding application **118** or the online concierge system **102** may generate (e.g., using the interface engine **328**) a set of augmented reality elements **715** that include a set of instructions for navigating the modified route **610** within the warehouse **150**. The wayfinding application **118** or the online concierge system **102** may then send (e.g., using the interface engine **328**) the set of augmented reality elements **715** to the display area of the client device **110**. Similarly, in embodiments in which the route **610** is modified and the wayfinding application **118** or the online concierge system **102** generates an augmented reality element **715** that includes the route **610**, the wayfinding application **118** or the online concierge system **102** may update (e.g., using the interface engine **328**) the augmented reality element **715**. Once the augmented reality element **715** is updated to reflect the modified route **610** (e.g., with an icon representing the location associated with the replacement item and any remaining target items, etc.), it may be sent (e.g., using the interface engine **328**) to the display area of the client device **110**.

[0146] Referring again to FIG. 5, if the wayfinding application **118** or the online concierge system **102** determines **535** that the target item is at the location within the warehouse **150** associated with the target item (or elsewhere within the warehouse **150**, in various embodiments), it may alert **555** the user of the client device **110** to the target item. It may do so by generating **560** (e.g., using the interface engine **328**) an augmented reality element **715** that calls attention to the target item and sending **565** (e.g., using the interface engine **328**) the augmented reality element **715** to the display area of the client device **110**. Similar to the augmented reality elements **715** described above, the augmented reality element **715** may call attention to the target item by highlighting it (e.g., with one or more colors) or with icons (e.g., arrows or circles), animations, etc. Once the wayfinding application **118** or the online concierge system **102** alerts (**540** or **555**) the user of the client device **110** to the replacement item or the target item, one or more steps of the flowchart may be repeated (e.g., by proceeding back to step **530**) for any remaining target items.

[0147] An example of the augmented reality element 715 that calls attention to the target item is shown in FIG. 7D, which illustrates a set of augmented reality elements 715 that calls attention to an item within a warehouse 150, according to one or more embodiments, and continues the example described above in conjunction with FIGS. 6, 7A, and 7B. In the example of FIG. 7D, the augmented reality element 715d that calls attention to the location of the target item detected 530 within the display area of the client device 110 is an icon with a number (1) that corresponds to the number of the target item within the shopping list included in augmented reality element 715b. In this example, once sent to the display area of the client device 110, the augmented reality element 715d hovers next to a particular portion of the display area in which a shelf corresponding to the location associated with the target item is detected 530. In the above example, once the wayfinding application 118 or the online concierge system 102 determines 535 that the target item is at the location within the warehouse 150 associated with the target item, the wayfinding application 118 or the online concierge system 102 alerts 555 the user to the target item by generating 560 an additional augmented reality element 715h that calls attention to the target item and sending 565 the additional augmented reality element 715h to the display area of the client device 110. Continuing with this example, the augmented reality element 715h that calls attention to the target item is a halo that is overlaid onto a portion of the display area of the client device 110 at which the target item is detected (e.g., by the item/location detection engine 330).

[0148] In various embodiments, as the user of the client device 110 travels within the warehouse 150, the wayfinding application 118 or the online concierge system 102 may detect (e.g., using the item/location detection engine 330) that the user has deviated from the route 610. The wayfinding application 118 or the online concierge system 102 may do so based on the route 610, the layout of the warehouse 150, and image or video data captured by the camera of the client device 110. For example, the wayfinding application 118 or the online concierge system 102 may detect the location of the client device 110 within the warehouse 150 based on the layout of the warehouse 150 and physical elements detected within the display area of the client device 110. In this example, the wayfinding application 118 or the online concierge system 102 may detect that the user of the client device 110 has deviated from the route 610 if the detected location does not correspond to a location within the warehouse 150 along the route 610.

[0149] In embodiments in which the wayfinding application 118 or the online concierge system 102 detects that the user of the client device 110 has deviated from the route 610, the wayfinding application 118 or the online concierge system 102 may modify (e.g., using the route engine 324) the route 610 based on a current location of the client device 110 within the warehouse 150. For example, if the user of the client device 110 deviates from the route 610 after collecting some of the target items, the wayfinding application 118 or the online concierge system 102 may detect that the user has deviated from the route 610 and determine a current location of the client device 110. In this example, the wayfinding application 118 or the online concierge system 102 may then modify the route 610 based on the current location of the client device 110 within the warehouse 150 and the locations associated with the remaining target items.

In this example, based on the modified route 610, the target items may be collected in an order that is different from that associated with the original route 610. In embodiments in which the route 610 is modified, the wayfinding application 118 or the online concierge system 102 may generate (e.g., using the interface engine 328) a set of augmented reality elements 715 including a set of instructions for navigating the modified route 610 within the warehouse 150 and send (e.g., using the interface engine 328) the set of augmented reality elements 715 to the display area of the client device 110.

[0150] In some embodiments, as the user of the client device 110 travels within the warehouse 150, the wayfinding application 118 or the online concierge system 102 also may detect (e.g., using the item/location detection engine 330) a quantity of an item at the warehouse 150. In such embodiments, the wayfinding application 118 or the online concierge system 102 may then update (e.g., using the inventory management engine 302) information describing the inventory of the item at the warehouse 150 (e.g., stored in the inventory database 304) to reflect the quantity of the item that was detected. Furthermore, in such embodiments, the wayfinding application 118 or the online concierge system 102 also or alternatively may update the layout of the warehouse 150 (e.g., stored in the layout database 322). In some embodiments, the layout of the warehouse 150 may be updated if at least a threshold quantity of an item is detected at the warehouse 150 and the wayfinding application 118 or the online concierge system 102 determines (e.g., using the item/location detection engine 330) that the threshold quantity of the item detected is not at a location within the warehouse 150 associated with the item. For example, suppose that the wayfinding application 118 or the online concierge system 102 detects at least ten cans of a particular brand and type of soup and determines that the cans of soup are not at a location within the warehouse 150 associated with the can of soup based on the layout of the warehouse 150. In this example, the wayfinding application 118 or the online concierge system 102 may update the layout of the warehouse 150 to reflect that the location within the warehouse 150 at which the cans of soup were detected is associated with the particular brand and type of soup.

[0151] Furthermore, as the user of the client device 110 travels within the warehouse 150, an augmented reality element 715 that includes the list of target items may be updated (e.g., via the interface engine 328) and sent (e.g., via the interface engine 328) to the display area of the client device 110. As described above, the wayfinding application 118 or the online concierge system 102 may update an augmented reality element 715 that includes the list of target items if a target item is unavailable or if a replacement item is identified. In various embodiments, the augmented reality element 715 also may be updated if a replacement item is unavailable, if a target item or a replacement item is collected, etc. An example of this is shown in FIG. 7E, which illustrates a set of augmented reality elements 715 for navigating a route 610 within a warehouse 150 to one or more locations associated with one or more items, according to one or more embodiments, and continues the example described above in conjunction with FIGS. 6 and 7A-7C. In the example of FIG. 7E, as the target items and the replacement item in the shopping list included in augmented reality element 715b are collected, augmented reality element 715b is updated by checking a checkbox next to each collected

item. In this example, the user of the client device **110** is approaching an augmented reality element **715i** that calls attention to a location of the fifth target item detected **530** within the display area of the client device **110**, which is an icon with a number (5) that corresponds to the number of the target item within the shopping list included in augmented reality element **715b**. Continuing with this example, checkboxes next to the fifth and sixth target items within the shopping list included in augmented reality element **715b** have not been checked because they have not yet been collected by the user of the client device **110**.

[0152] In embodiments in which an augmented reality element **715** that includes the list of target item(s) is updated, the augmented reality element **715** may be updated upon detecting (e.g., using the item/location detection engine **330**) that an item has been collected or in response to receiving an input from the user via the interface described above. For example, if the wayfinding application **118** or the online concierge system **102** detects a target item and a shopping basket being held by the user of the client device **110**, the wayfinding application **118** or the online concierge system **102** also may detect that the target item is within the shopping basket. In this example, an augmented reality element **715** that includes the target item in a shopping list may be updated by checking a checkbox next to the target item. As an additional example, as the replacement item and target items in the shopping list included in augmented reality element **715b** shown in FIG. 7E are collected, the wayfinding application **118** or the online concierge system **102** may receive requests from the user of the client device **110** to check the collected items off the list via the interface described above. In this example, augmented reality element **715b** may be updated in response to each request by checking off the collected item.

[0153] In embodiments in which the wayfinding application **118** or the online concierge system **102** generates an augmented reality element **715** that includes the route **610**, the augmented reality element **715** also may be updated (e.g., as the user of the client device **110** moves within the warehouse **150**, as the target item(s) is/are collected, as the route **610** is modified, etc.). In the examples of FIGS. 7A-7E, as the user of the client device **110** moves along the route **610**, the icon representing the location of the client device **110** also moves, tracking the location of the client device **110** within the warehouse **150**. In this example, as the target items are collected, icons representing the locations associated with the target items are removed.

[0154] In some embodiments, the wayfinding application **118** or the online concierge system **102** also may generate (e.g., using the interface engine **328**) one or more augmented reality elements **715** configured to perform various functions, which is/are then sent (e.g., using the interface engine **328**) to the display area of the client device **110**. In such embodiments, the augmented reality element(s) **715** may be configured to perform the functions with respect to items included among the inventory of the warehouse **150** detected within the display area of the client device **110**, including calling attention to them and/or their attributes, obscuring them, comparing them, recommending different items, and promoting them. In some embodiments, augmented reality elements **715** also or alternatively may be configured to perform the same functions with respect to other physical elements (e.g., advertisements, coupons, etc.) detected (e.g., by the item/location detection engine **330**) within the display

area of the client device **110**. In various embodiments, the user of the client device **110** may adjust one or more settings to change an augmented reality element **715** (e.g., its appearance, how it performs a function, etc.) via the interface described above.

[0155] The additional augmented reality element(s) **715** may be configured to perform different functions for various reasons. In addition to calling attention to target items and replacement items detected within the display area of the client device **110**, an augmented reality element **715** also may call attention to an item detected within the display area of the client device **110** if one or more attributes of the item match profile information associated with the customer **204**. An augmented reality element **715** may call attention to an attribute of an item detected within the display area of the client device **110** if the item having the attribute is a target item, if the attribute matches the profile information associated with the customer **204**, etc. An augmented reality element **715** may obscure an item detected within the display area of the client device **110** if the profile information associated with the customer **204** indicates that the customer **204** dislikes the item/attributes of the item, if the profile information indicates the customer **204** has a restriction, such as a dietary restriction, associated with the item, etc. An augmented reality element **715** may compare two or more items detected within the display area of the client device **110** if the user of the client device **110** is holding the items, if the items are detected within the display area for at least a threshold amount of time, etc. In addition to recommending a replacement item for a target item that is out of stock, an augmented reality element **715** may recommend an item detected within the display area of the client device **110** if the customer **204** is likely to purchase it (e.g., based on their order/purchase history or other profile information, other items included among the target item(s) etc.), if it is a replacement item the customer **204** is likely to prefer to a target item, etc. An augmented reality element **715** may promote an item detected within the display area of the client device **110** if it is a new item, if the customer **204** is considering purchasing it, if it is on sale or associated with a promotion, etc.

[0156] In embodiments in which the wayfinding application **118** or the online concierge system **102** generates the augmented reality element(s) **715** configured to perform one or more of the functions described above, the augmented reality element(s) **715** may be generated based on various types of information. Examples of such information include inferences made by the wayfinding application **118** or the online concierge system **102** (e.g., using the experience enhancement engine **338**) and information describing replacement items identified by the wayfinding application **118** or the online concierge system **102** (e.g., using the item replacement engine **332** and the experience enhancement engine **338**). For example, if the wayfinding application **118** or the online concierge system **102** detects that the user of the client device **110** is holding two items, it may infer that the user is comparing the items and may generate an augmented reality element **715** that compares various attributes of the items (e.g., their sugar contents, calorie counts, etc.). Additional examples of information that the wayfinding application **118** or the online concierge system **102** may use to generate the additional augmented reality element(s) **715** include information describing profile or other information associated with the customer **204** that the wayfinding

application **118** or the online concierge system **102** matches (e.g., using the experience enhancement engine **338**) with attributes of items and information accessed from various databases (e.g., the inventory database **304**, the customer database **314**, the content database **334**, etc.). For example, if the profile information associated with the customer **204** indicates that the customer **204** is on a low-sugar diet, the wayfinding application **118** or the online concierge system **102** may match this information with an attribute of an item the user of the client device **110** is holding corresponding to its low sugar content. In this example, based on the information describing the match, an augmented reality element **715** generated by the wayfinding application **118** or the online concierge system **102** calling attention to an attribute of the item may highlight its sugar content.

[0157] In some embodiments, the wayfinding application **118** or the online concierge system **102** may store information describing augmented reality elements **715** sent to the client device **110** (e.g., in the customer database **314**), such that this information is included among the profile information associated with the customer **204**. For example, suppose that the wayfinding application **118** or the online concierge system **102** sends an augmented reality element **715** to the display area of the client device **110**, in which the augmented reality element **715** promotes an item and includes a content item (e.g., a coupon) associated with the item. In this example, information describing the augmented reality element **715** (e.g., information indicating that it promoted the item, information identifying the item and the content item, etc.) and a date and time that it was sent to the display area of the client device **110** may be stored among the profile information associated with the customer **204**.

Additional Considerations

[0158] The foregoing description of the embodiments of the invention has been presented for the purpose of illustration; it is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in light of the above disclosure.

[0159] Some portions of this description describe the embodiments of the invention in terms of algorithms and symbolic representations of operations on information. These algorithmic descriptions and representations are commonly used in the data processing arts to convey the substance of their work effectively to others. These operations, while described functionally, computationally, or logically, are understood to be implemented by computer programs or equivalent electrical circuits, microcode, or the like. Furthermore, it has also proven convenient at times, to refer to these arrangements of operations as modules, without loss of generality. The described operations and their associated modules may be embodied in software, firmware, hardware, or any combinations thereof.

[0160] Any of the steps, operations, or processes described herein may be performed or implemented with one or more hardware or software modules, alone or in combination with other devices. In one embodiment, a software module is implemented with a computer program product comprising a computer-readable medium containing computer program code, which can be executed by a computer processor for performing any or all of the steps, operations, or processes described.

[0161] Embodiments of the invention may also relate to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes, and/or it may comprise a computing device selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a tangible computer readable storage medium, which include any type of tangible media suitable for storing electronic instructions and coupled to a computer system bus. Furthermore, any computing systems referred to in the specification may include a single processor or may be architectures employing multiple processor designs for increased computing capability.

[0162] Embodiments of the invention may also relate to a computer data signal embodied in a carrier wave, where the computer data signal includes any embodiment of a computer program product or other data combination described herein. The computer data signal is a product that is presented in a tangible medium or carrier wave and modulated or otherwise encoded in the carrier wave, which is tangible, and transmitted according to any suitable transmission method.

[0163] Finally, the language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the inventive subject matter. It is therefore intended that the scope of the invention be limited not by this detailed description, but rather by any claims that issue on an application based hereon. Accordingly, the disclosure of the embodiments of the invention is intended to be illustrative, but not limiting, of the scope of the invention, which is set forth in the following claims.

What is claimed is:

1. A method comprising:

capturing, by a camera of a client device within a warehouse, an image depicting a plurality of structural elements within the warehouse;

applying an item detection engine to the captured image to identify the set of structural elements within the image, wherein the item detection engine comprises a machine-learning model that is configured to identify elements within images captured by client devices within warehouses;

accessing, from a layout database, a layout of the warehouse, wherein the layout describes a set of locations within the warehouse associated with structural elements within the warehouse;

generating a predicted location of the client device based on the accessed layout of the warehouse and the set of structural elements identified in the image;

applying the item detection engine to the captured image and the predicted location of the client device to identify a set of items depicted in the image;

generating, by an AR interface engine, a set of augmented reality elements that call attention to the set of items identified in the image data; and

sending the set of augmented reality elements to the client device for display to the user.

2. The method of claim 1, wherein the set of structural elements comprises one or more of aisles, shelves, display areas, entrances, exits, floors, ceilings, walls, stairs, or elevators.

3. The method of claim 1, further comprising determining an orientation of the client device within the warehouse based on the identified set of structural elements and the layout of the warehouse.

4. The method of claim 1, wherein the set of augmented reality elements comprises one or more of highlights, icons, arrows, circles, or animations to call attention to the set of items.

5. The method of claim 1, wherein the set of augmented reality elements obscures one or more items in the image by blocking, blurring, or otherwise visually modifying the display of the item.

6. The method of claim 1, wherein the set of augmented reality elements includes a comparison of two or more items, of the identified set of items.

7. The method of claim 1, wherein the machine-learning model is trained using images of physical elements stored in an inventory database, a layout database, or a content database.

8. The method of claim 1, wherein the set of augmented reality elements is generated based on a set of preferences or dietary restrictions associated with a user profile.

9. The method of claim 1, wherein generating the prediction location for the client device comprises:

accessing location sensor data captured by a location sensor of the client device.

10. A non-transitory computer-readable medium storing instructions that, when executed by a computing system, causes the computing system to perform operations comprising:

capturing, by a camera of a client device within a warehouse, an image depicting a plurality of structural elements within the warehouse;

applying an item detection engine to the captured image to identify the set of structural elements within the image, wherein the item detection engine comprises a machine-learning model that is configured to identify elements within images captured by client devices within warehouses;

accessing, from a layout database, a layout of the warehouse, wherein the layout describes a set of locations within the warehouse associated with structural elements within the warehouse;

generating a predicted location of the client device based on the accessed layout of the warehouse and the set of structural elements identified in the image;

applying the item detection engine to the captured image and the predicted location of the client device to identify a set of items depicted in the image;

generating, by an AR interface engine, a set of augmented reality elements that call attention to the set of items identified in the image data; and

sending the set of augmented reality elements to the client device for display to the user.

11. The computer-readable medium of claim 10, wherein the set of structural elements comprises one or more of aisles, shelves, display areas, entrances, exits, floors, ceilings, walls, stairs, or elevators.

12. The computer-readable medium of claim 10, further comprising determining an orientation of the client device within the warehouse based on the identified set of structural elements and the layout of the warehouse.

13. The computer-readable medium of claim 10, wherein the set of augmented reality elements comprises one or more of highlights, icons, arrows, circles, or animations to call attention to the set of items.

14. The computer-readable medium of claim 10, wherein the set of augmented reality elements obscures one or more items in the image by blocking, blurring, or otherwise visually modifying the display of the item.

15. The computer-readable medium of claim 10, wherein the set of augmented reality elements includes a comparison of two or more items, of the identified set of items.

16. The computer-readable medium of claim 10, wherein the machine-learning model is trained using images of physical elements stored in an inventory database, a layout database, or a content database.

17. The computer-readable medium of claim 10, wherein the set of augmented reality elements is generated based on a set of preferences or dietary restrictions associated with a user profile.

18. The computer-readable medium of claim 10, wherein generating the prediction location for the client device comprises:

accessing location sensor data captured by a location sensor of the client device.

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